

PRACTICE SHEET-1

1. In circular measure, the value of the angle $11^\circ 15'$ is
 (1) $\frac{\pi}{16}$ (2) $\frac{\pi}{8}$
 (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{32}$
2. In a triangle ABC, $\angle ABC = 75^\circ$ and $\angle ACB = \frac{\pi}{4}$. The circular measure of $\angle BAC$ is
 (1) $\frac{5\pi}{12}$ radian (2) $\frac{\pi}{3}$ radian
 (3) $\frac{\pi}{6}$ radian (4) $\frac{\pi}{2}$ radian
3. The circular measure of an angle of an isosceles triangle is $\frac{5\pi}{9}$. Circular measure of one of the other angles must be:
 (1) $\frac{5\pi}{18}$ (2) $\frac{5\pi}{9}$
 (3) $\frac{2\pi}{9}$ (4) $\frac{4\pi}{9}$
4. The degree measure of 1 radian (taking $\pi = 22/7$) is
 (1) $57^\circ 61' 22''$ (approx.)
 (2) $57^\circ 16' 22''$ (approx.)
 (3) $57^\circ 22' 16''$ (approx.)
 (4) $57^\circ 32' 16''$ (approx.)
5. Consider the following statements
 I. 1° in radian measure is less than 0.03 radians.
 II. 1 radian in degree measure is greater than 45°
 Which of the above statements is/are correct?
 (a) Only II (b) Only I
 (c) Both I and II (d) Neither 1 nor 2
6. If the sum of two angles is 135° and their difference is $\frac{\pi}{12}$, then the circular measure of the greater angle is
 (1) $\frac{2\pi}{3}$ (2) $\frac{3\pi}{5}$
 (3) $\frac{5\pi}{12}$ (4) $\frac{\pi}{3}$
7. If the sum and difference of two angles are $22/9$ radian and 36° respectively, then the value of smaller angle in degree taking the value of π as $22/7$ is:
 (1) 52° (2) 60°
 (3) 56° (4) 48°
8. The circular measure of the included angle formed by the hour hand and minute hand of a clock at 3 p.m. will be:
 (1) $\frac{\pi}{4}$ (2) $\frac{\pi}{3}$
 (3) $\frac{5\pi}{12}$ (4) $\frac{\pi}{2}$
9. What is the measure of central angle of the arc whose length is 11 cm and radius of the circle is 14 cm?
 (1) 45° (2) 60°
 (3) 75° (4) 90°
10. The radian measure of 1 degree (taking $\pi = 22/7$) is
 (1) 0.016° (2) 0.017°
 (3) 0.018° (4) 0.019°
11. What is the angle between the hands of clock when time is 7:20pm?
 (1) 100° (2) 120°
 (3) 108° (4) 90°
12. What is the angle between the hands of clock when time is 1:40pm?
 (1) 150° (2) 120°
 (3) 120° (4) 170°
13. What is the angle between the hands of clock when time is 3:30am?
 (1) 90° (2) 80°
 (3) 75° (4) 60°
14. At what time between 2 and 3 o'clock will the hands of a clock be together?
 (1) $10\frac{10}{11}$ min, past 2
 (2) 10 min, past 2
 (3) $20\frac{10}{11}$ min, past 2
 (4) 12min, past 2

15. How many times do the hands of a clock coincide in a day?
 (1) 20 (2) 21
 (3) 22 (4) 24
16. In circular measure, the value of the angle $22^{\circ}30'$ is
 (1) $\frac{\pi}{16}$ (2) $\frac{\pi}{8}$
 (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{32}$
17. In circular measure, the value of the angle $9^{\circ}22'30''$ is
 (1) $\frac{7\pi}{96}$ (2) $\frac{5\pi}{96}$
 (3) $\frac{11\pi}{48}$ (4) $\frac{5\pi}{48}$
18. In degree measure, the value of the angle $\frac{\pi}{16}$ is
 (1) $22^{\circ}30'$ (2) $11^{\circ}15'$
 (3) $30^{\circ}30'$ (4) $15^{\circ}30'$
19. In degree measure, the value of the angle $\frac{2\pi}{5}$ is
 (1) 72° (2) 45°
 (3) 36° (4) 20°
20. In degree measure, the value of the angle $\frac{3\pi}{8}$ is
 (1) $72^{\circ}30'$ (2) 45°
 (3) $67^{\circ}30'$ (4) 20°
21. Between 6 and 7 O' clock, both hour hand and minute hand will coincide past 6 O'clock between
 (1) 31-32 minutes (2) 32-33 minutes
 (3) 33-34 minutes (4) 34-35 minutes
22. A clock is started at noon. By 10 minutes past 5, the hour hand has turned through:
 (1) 145° (2) 150°
 (3) 155° (4) 160°
23. How many times are the hands of clock at right angle in a day?
 (1) 22 (2) 24
 (3) 44 (4) 48
24. A clock gains 20 seconds for every 3 hours of time. If a clock is set at a correct time of 2 am on Friday, what would it indicate at 6:30 pm, Saturday?
 (1) 6.32.00pm (2) 6.32.46pm
 (3) 6.34.30pm (4) 6.38.56 pm
25. A person goes to a market between 7pm and 8pm. When he comes back, he finds that the hour hand and the minute hand of the clock have interchanged their positions. For how much time (approximately) was he out of his house?
 (a) 55.38 minutes (b) 55.48 minutes
 (c) 55.57 minutes (d) 55.67 minutes
26. Find the time between 8am and 9am when the hours and minutes hand would form a 90 degree between themselves?
 (1) $25\frac{7}{11}$ minutes past 8 am
 (2) $27\frac{3}{11}$ minutes past 8 am
 (3) $29\frac{8}{11}$ minutes past 8 am
 (4) $24\frac{5}{11}$ minutes past 8 am
27. A clock strikes once 1 O'clock, twice at 2 O'clock and thrice at 3 O'clock and so on. If it takes 8 seconds to strike at 5 O'clock, the time taken by it to strike at 10 O'clock is:
 (a) 14 seconds
 (b) 16 seconds
 (c) 18 seconds
 (d) None of the above
28. The degree measure of 3 radian (taking $\pi = \frac{22}{7}$) is
 (1) $171^{\circ}47'66''$ (approx.)
 (2) $171^{\circ}48'06''$ (approx.)
 (3) $171^{\circ}49'66''$ (approx.)
 (4) $171^{\circ}49'06''$ (approx.)
29. Consider the following:
 1. $57^{\circ} > 1^{\circ}$ 2. $1^{\circ} < 0.02^{\circ}$ 3. $45^{\circ} < 1^{\circ}$
 Which of the above are correct?
 (a) 1 and 2 only (b) 2 and 3 only
 (c) 1 and 3 only (d) 1, 2 and 3
30. In circular measure, the value of the angle $5^{\circ}37'30''$ is
 (1) $\frac{\pi}{16}$ (2) $\frac{\pi}{8}$
 (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{32}$

ANSWER KEY

1.	(1)	2.	(2)	3.	(3)	4.	(2)	5.	(3)	6.	(3)	7.	(1)	8.	(4)	9.	(1)	10.	(2)
11.	(1)	12.	(4)	13.	(3)	14.	(1)	15.	(3)	16.	(2)	17.	(2)	18.	(2)	19.	(1)	20.	(3)
21.	(2)	22.	(3)	23.	(3)	24.	(3)	25.	(a)	26.	(2)	27.	(3)	28.	(4)	29.	(2)	30.	(4)

PRACTICE SHEET-2

- If $3\tan\theta = 4$, then $\sqrt{\frac{1-\sin\theta}{1+\sin\theta}}$ is equal to
 (1) $1/2$ (2) $2/3$
 (3) $1/3$ (4) none of these
- What is the value of $\log(\tan 1^\circ) + \log(\tan 2^\circ) + \log(\tan 3^\circ) + \dots + \log(\tan 89^\circ)$ equal to
 (1) 0 (2) 1
 (3) 2 (4) -1
- If $\sec\theta = \frac{13}{5}$, then $\frac{2\sin\theta - 3\cos\theta}{4\sin\theta - 9\cos\theta}$ is equal to
 (1) $1/3$ (2) -3
 (3) 3 (4) $\sqrt{3}$
- If $\sin\theta = \frac{a^2 - b^2}{a^2 + b^2}$, then $\tan\theta$ is equal to
 (1) $\frac{a^2 + b^2}{2ab}$ (2) $\frac{2ab}{a^2 - b^2}$
 (3) $\frac{a^2 - b^2}{2ab}$ (4) $\frac{ab}{a^2 + b^2}$
- If $\cos\theta = 0.96$, then $\frac{1}{\sin\theta} + \frac{1}{\tan\theta}$ is equal to?
 (1) 0.98 (2) 3
 (3) 4 (4) 7
- If $0 < x < 45^\circ$ and $45^\circ < y < 90^\circ$ then which one of the following is correct?
 (1) $\sin x = \sin y$ (2) $\sin x < \sin y$
 (3) $\sin x > \sin y$ (4) $\sin x \leq \sin y$
- What is the value of $[(1 - \sin^2\theta) \sec^2\theta + \tan^2\theta](\cos^2\theta + 1)$ when $0 < \theta < 90^\circ$?
 (1) 2 (2) > 2
 (3) ≥ 2 (4) < 2
- If $7\cos^2\theta + 3\sin^2\theta = 4$, $0 < \theta < \frac{\pi}{2}$, then what is the value of $\tan\theta$?
 (1) $\sqrt{7}$ (2) $7/3$
 (3) 3 (4) $\sqrt{3}$
- If $0 \leq \theta \leq \frac{\pi}{2}$ and $p = \sec^2\theta$, then which one of the following is correct?
 (1) $p < 1$ (2) $p = 1$
 (3) $p > 1$ (4) $p \geq 1$
- What is the value of $\cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 90^\circ$?
 (1) $1/2$ (2) 0
 (3) 1 (4) 2
- What is $\cot 15^\circ \cot 20^\circ \cot 70^\circ \cot 75^\circ$ equal to?
 (1) -1 (2) 0
 (3) 1 (4) 2
- If $\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} = 2$, with $0 < \theta < 90^\circ$, then what is θ equal to?
 (1) 30° (2) 45°
 (3) 60° (4) 75°
- The smallest side of a right angled triangle has length 2 cm. The tangent of one acute angle is $3/4$. What is the hypotenuse of the triangle?
 (1) 5 cm (2) 2.5 cm
 (3) 1.25 cm (4) $10/3$ cm
- Which one of the following statements is true in respect of the expression $\sin 31^\circ + \sin 32^\circ$?
 (1) Its value is 0
 (2) Its value is 1
 (3) Its value is less than 1
 (4) Its value is greater than 1.
- If $\alpha + \beta = \frac{\pi}{2}$ and $\sin\alpha = \frac{1}{2}$, then β is
 (1) 30° (2) 60°
 (3) 45° (4) $22\frac{1}{2}^\circ$

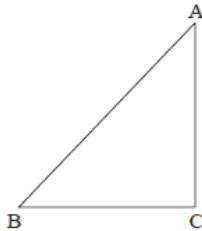
16. Consider the following statements
I. 1° in radian measure is less than 0.03 radians.
II. 1 radian in degree measure is greater than 45°
Which of the above statements is/are correct?
(1) Only II (2) Only I
(3) Both I and II (4) Neither 1 nor 2
- Direction (for next three)**
- Let $\sin(A + B) = 1$ and $\sin(A - B) = \frac{1}{2}$,
where $A, B = \frac{1}{2}$, where $A, B \in \left[0, \frac{\pi}{2}\right]$
17. What is the value of A?
(1) 30° (2) 60°
(3) 45° (4) $22\frac{1}{2}^\circ$
18. What is the value of $\tan(A + 2B) \tan(2A + B)$?
(1) -1 (2) 0
(3) 1 (4) 2
19. What is the value of $\sin^2 A - \sin^2 B$?
(1) 0 (2) $1/2$
(3) 1 (4) 2
20. The value of $\cos 25^\circ - \sin 25^\circ$ is
(1) Positive but less than 1
(2) Positive but greater than 1
(3) Negative
(4) 0
21. If $\tan \theta = 2$, then the value of $\frac{\sin \theta}{\sin^3 \theta + \cos^3 \theta}$ is
(1) $2/9$ (2) $5/9$
(3) $10/9$ (4) $50/9$
22. If $\sin \theta + \cos \theta = 1$, then $\sin \theta \cdot \cos \theta$ is equal to
(1) 0 (2) 1
(3) $1/2$ (4) $-1/2$
23. If $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = 3$, then the value of $\sin^4 \theta - \cos^4 \theta$ is
(1) $4/3$ (2) $3/4$
(3) $5/3$ (4) $3/5$
24. The value of $\sec^2 17^\circ - \frac{1}{\tan^2 73^\circ} - \sin 17^\circ \sec 73^\circ$ is
(1) 1 (2) 0
(3) -1 (4) 2
25. If $x = a \cos \theta \cos \alpha$, $y = a \cos \theta \sin \alpha$ and $z = a \sin \theta$, then the value of $(x^2 + y^2 + z^2)$ is
(1) $2a^2$ (2) $4a^2$
(3) $9a^2$ (4) a^2
26. Which one of the following is correct?
(a) There is only one θ with $0^\circ < \theta < 90^\circ$ such that $\sin \theta = a$, where a is a real number.
(b) There is more than one θ with $0^\circ < \theta < 90^\circ$ such that $\sin \theta = a$, where a is a real number.
(c) There is no θ with $0^\circ < \theta < 90^\circ$ such that $\sin \theta = a$, where a is a real number.
(d) There are exactly θ 's with $0^\circ < \theta < 90^\circ$ such that $\sin \theta = a$, where a is a real number
27. If $\tan \theta = \tan 30^\circ \cdot \tan 60^\circ$ and θ is an acute angle, then 2θ is equal to
(1) 30° (2) 45°
(3) 90° (4) 0°
28. The value of $(1 + \tan^2 \theta)(1 - \sin^2 \theta)$ is
(1) 2 (2) 1
(3) -1 (4) -2
29. If $r \sin \theta = 1$, $r \cos \theta = \sqrt{3}$ then the value of $(r^2 \tan \theta)$ is
(1) 4 (2) $1/\sqrt{3}$
(3) $4/\sqrt{3}$ (4) $4\sqrt{3}$
30. If $\sin \theta = \sqrt{3}/2$ and $0^\circ < \theta < 90^\circ$, then the value of $\tan(\theta - 15^\circ)$ is
(1) 1 (2) $\sqrt{3}$
(3) $1/\sqrt{3}$ (4) $\sqrt{2}$

ANSWER KEY

1.	(3)	2.	(1)	3.	(3)	4.	(3)	5.	(4)	6.	(2)	7.	(2)	8.	(4)	9.	(4)	10.	(2)
11.	(3)	12.	(2)	13.	(4)	14.	(4)	15.	(2)	16.	(3)	17.	(2)	18.	(3)	19.	(2)	20.	(1)
21.	(3)	22.	(1)	23.	(4)	24.	(2)	25.	(4)	26.	(1)	27.	(3)	28.	(2)	29.	(3)	30.	(1)

PRACTICE SHEET-3

1. What is $\frac{\sin^6 \theta - \cos^6 \theta}{\sin^2 \theta - \cos^2 \theta}$ equal to?
 - (1) $\sin^4 \theta - \cos^4 \theta$ (2) $1 - \sin^2 \theta \cos^2 \theta$
 - (3) $1 + \sin^2 \theta \cos^2 \theta$ (4) $1 - 3\sin^2 \theta \cos^2 \theta$
 2. If $\sin \theta + \cos \theta = 1$, then what is the value of $\sin \theta \cos \theta$?
 - (1) 2 (2) 0
 - (3) 1 (4) $1/2$
 3. What is $\sqrt{\frac{1+\sin \theta}{1-\sin \theta}}$ equal to?
 - (1) $\sec \theta - \tan \theta$ (2) $\sec \theta + \tan \theta$
 - (3) $\operatorname{cosec} \theta + \cot \theta$ (4) $\operatorname{cosec} \theta - \cot \theta$
 4. $\sin^2 \theta - 3 \sin \theta + 2 = 0$ will be true if
 - (1) $0 \leq \theta < 90$ (2) $0 < \theta < 90$
 - (3) $\theta = 0^\circ$ (4) $\theta = 90^\circ$
 5. In a $\triangle ABC$, $\angle ABC = 90^\circ$, $\angle ACB = 30^\circ$, $AB = 5$ cm. What is the length of AC ?
 - (1) 10 cm (2) 5 cm
 - (3) $5\sqrt{2}$ cm (4) $5\sqrt{3}$ cm
 6. If $7 \cos^2 \theta + 3 \sin^2 \theta = 4$ and $0 < \theta < 90^\circ$, then what is the value of $\tan \theta$?
 - (a) $\sqrt{7}$ (b) $7/3$
 - (c) 3 (d) $\sqrt{3}$
 7. For what value of θ is $(\sin \theta + \operatorname{cosec} \theta) = 2.5$, where $0 < \theta < 90^\circ$?
 - (1) 30° (2) 45°
 - (3) 60° (4) 90°
 8. $\frac{\cos \theta}{1-\sin \theta} - \frac{\cos \theta}{1+\sin \theta} = 2$, is satisfied by which of the following value of θ ?
 - (1) $\frac{\pi}{2}$ (2) $\frac{\pi}{3}$
 - (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{6}$
 9. Consider the following:
 1. $\frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta} = \cos^2 \theta (1 + \tan \theta)(1 - \tan \theta)$
 2. $\frac{1 + \sin \theta}{1 - \sin \theta} = (\tan \theta + \sec \theta)^2$
- Which of the above are correct?
- (1) only 1 (2) only 2
 - (3) Both 1 and 2 (4) Neither 1 nor 2
10. If x is the angle of first quadrant such that $\operatorname{cosec}^4 x = 17 + \cot^4 x$, then what is the value of $\sin x$?
 - (1) $1/3$ (2) $1/4$
 - (3) $1/9$ (4) $1/16$
 11. Consider the following equations:
 - I. $\operatorname{cosec}^2 x + \sec^2 x = \operatorname{cosec}^2 x \sec^2 x$
 - II. $\sec^2 x + \tan^2 x = \sec^2 x \tan^2 x$
 - III. $\operatorname{cosec}^2 x + \tan^2 x = \cot^2 x + \sec^2 x$
 Which of the above statements are correct?
 - (1) I and II (2) II and III
 - (3) I and III (4) All of these
 12. If the given figure, $BC = 15$ cm and $\sin B = 4/5$. What is the value of AB ?



 - (1) 25 cm (2) 20 cm
 - (3) 5 cm (4) 4 cm
 13. If $\sin^2 60^\circ + \cos^2 (3x - 9^\circ) = 1$, then what is the value of x ?
 - (1) 24° (2) 23°
 - (3) 22° (4) 21°
 14. What is the expression: $(\sin^4 x - \cos^4 x + 1) \operatorname{cosec}^2 x$ equal to?
 - (1) 1 (2) 2
 - (3) 0 (4) -1
 15. Assertion (A): $\tan 50^\circ > 1$.
Reason (R): $\tan \theta > 1$ for $0^\circ < \theta < 90^\circ$.
 - (1) A and R are correct and R is correct explanation of A.
 - (2) A and R are correct but R is not correct explanation of A.
 - (3) A is true but R is false.
 - (4) A is true but R is false.

16. Which one of the following is correct?
 (1) $\sin 35^\circ > \cos 55^\circ$ (2) $\cos 61^\circ > 1/2$
 (3) $\sin 32^\circ > 1/2$ (4) $\tan 44^\circ > 1$
17. Which of the following expression for $0^\circ < \theta < 90^\circ$ is/are independent of θ ?
 I. $\cos\theta(1 - \sin\theta)^{-1} + \cos\theta(1 + \sin\theta)^{-1}$
 II. $\cos\theta(1 + \operatorname{cosec}\theta)^{-1} + \cos\theta(\operatorname{cosec}\theta - 1)^{-1}$
 Select the correct answer using the codes given below:
 (1) Only I (2) Only II
 (3) Both I and II (4) Neither I nor II
18. If $p = \sin^{10} x$, then which one of the following is correct for any value x ?
 (1) $p \geq 1$ (2) $0 \leq p \leq 1$
 (3) $1 \leq p \leq 2$ (4) None of these
19. If $\sin(x - y) = 1/2$ and $\cos(x + y) = 1/2$, then what is the value of x ?
 (1) 15° (2) 30°
 (3) 45° (4) 60°
20. If $2x^2 \cos 60^\circ - 4 \cot^2 45^\circ - 2 \tan 60^\circ = 0$, then what is the value of x ?
 (1) 2 (2) 3
 (3) $\sqrt{3} - 1$ (4) $\sqrt{3} + 1$
21. Under which one of the following conditions is the trigonometrically identity $\frac{\sin x}{(1 + \cos x)} = \frac{(1 - \cos x)}{\sin x}$ true?
 (1) x is not a multiple of 360°
 (2) x is not an odd multiple of 180°
 (3) x is not a multiple of 180°
 (4) None of the above
22. If A, B, C and D are the successive angles of a cyclic quadrilateral, then what is $\cos A + \cos B + \cos C + \cos D$ are equal to?
 (1) 4 (2) 2
 (3) 1 (4) 0
23. If $x = \operatorname{cosec} \theta - \sin \theta$ and $y = \sec \theta - \cos \theta$, then the relation between x and y is
 (1) $x^2 + y^2 + 3 = 1$
 (2) $x^2 y^2 (x^2 + y^2 + 3) = 1$
 (3) $x^2 (x^2 + y^2 - 5) = 1$
 (4) $y^2 (x^2 + y^2 - 5) = 1$
24. If $2y \cos \theta = x \sin \theta$ and $2x \sec \theta - y \operatorname{cosec} \theta = 3$ then what is the value of $(x^2 + 4y^2)$?
 (1) 4 (2) 1
 (3) 2 (4) 5
25. If $\cos \theta + \sin \theta = m$ and $\sec \theta + \operatorname{cosec} \theta = n$ then the value of $n(m^2 - 1)$ is equal to :
 (1) $2m$ (2) mn
 (3) $4mn$ (4) $2n$
26. If x is an acute angle and $2 \sin x + 15 \cos^2 x = 7$ then the value of $\cot x$ is
 (1) $4/3$ (2) $4/5$
 (3) $5/4$ (4) $3/4$
27. If $0^\circ < A < 90^\circ$, then the value of $\tan^2 A + \cot^2 A - \sec^2 A \operatorname{cosec}^2 A$ is
 (1) 0 (2) 1
 (3) 2 (4) -2
28. If $\sec \theta + \tan \theta = 2$, then the value of $\sin \theta$ is (assume that $0 < \theta < 90^\circ$)
 (1) 0.4 (2) 0.5
 (3) 0.6 (4) 0.8
29. The value of $(1 + \cot \theta - \operatorname{cosec} \theta)(1 + \tan \theta + \sec \theta)$ is equal to
 (1) 1 (2) 2
 (3) 0 (4) -1
30. $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 + \tan \theta} =$
 (1) $1 - \tan \theta - \cot \theta$ (2) $1 + \tan \theta - \cot \theta$
 (3) $1 - \tan \theta + \cot \theta$ (4) $1 + \tan \theta + \cot \theta$

ANSWER KEY

1.	(2)	2.	(2)	3.	(2)	4.	(4)	5.	(1)	6.	(4)	7.	(1)	8.	(3)	9.	(3)	10.	(1)
11.	(3)	12.	(1)	13.	(2)	14.	(2)	15.	(3)	16.	(3)	17.	(4)	18.	(2)	19.	(2)	20.	(4)
21.	(3)	22.	(2)	23.	(2)	24.	(1)	25.	(1)	26.	(4)	27.	(4)	28.	(3)	29.	(2)	30.	(2)

PRACTICE SHEET-4

1. The value of $\sin^2 1^\circ + \sin^2 5^\circ + \sin^2 9^\circ + \dots + \sin^2 89^\circ$ is
 (1) $11\frac{1}{2}$ (2) $11\sqrt{2}$
 (3) 11 (4) $11/\sqrt{2}$
2. The numerical value of $\cot 18^\circ \left(\cot 72^\circ \cos^2 22^\circ + \frac{1}{\tan 72^\circ \sec^2 68^\circ} \right)$ is
 (1) 1 (2) $\sqrt{2}$
 (3) 3 (4) $1/\sqrt{3}$
3. If $\angle A$ and $\angle B$ are complementary to each other, then the value of $\sec^2 A + \sec^2 B - \sec^2 A \cdot \sec^2 B$ is
 (1) 1 (2) -1
 (3) 2 (4) 0
4. $\sin^2 5^\circ + \sin^2 6^\circ + \dots + \sin^2 84^\circ + \sin^2 85^\circ = ?$
 (1) $39\frac{1}{2}$ (2) $40\frac{1}{2}$
 (3) 40 (4) $39\frac{1}{\sqrt{2}}$
5. The value of $\frac{\sin 39^\circ}{\cos 51^\circ} + 2 \tan 11^\circ \tan 31^\circ \tan 45^\circ$
 $\tan 59^\circ \tan 79^\circ - 3(\sin^2 21^\circ + \sin^2 69^\circ)$ is:
 (1) 2 (2) -1
 (3) 1 (4) 0
6. If $\frac{\cos^2 \theta}{\cot^2 \theta - \cos^2 \theta} = 3$ and $0^\circ < \theta < 90^\circ$, then the value of θ is
 (1) 30° (2) 45°
 (3) 60° (4) None of these
7. If $A = \tan 11^\circ \tan 29^\circ$, $B = 2 \cot 61^\circ \cot 79^\circ$, then:
 (1) $A = 2B$ (2) $A = -2B$
 (3) $2A = B$ (4) $2A = -B$
8. If $0^\circ < \theta < 90^\circ$, the value of $\sin \theta + \cos \theta$ is
 (1) equal to 1 (2) greater than 1
 (3) less than 1 (4) equal to 2
9. The value of $\cot \theta \cdot \tan(90^\circ - \theta) - \sec(90^\circ - \theta) \operatorname{cosec} \theta + (\sin^2 25^\circ + \sin^2 65^\circ) + \sqrt{3} (\tan 5^\circ \tan 15^\circ \cdot \tan 30^\circ \cdot \tan 75^\circ \cdot \tan 85^\circ)$ is :
 (1) 1 (2) -1
 (3) 2 (4) 0
10. If $\tan\left(\frac{\pi}{2} - \frac{\theta}{2}\right) = \sqrt{3}$, then the value of $\cos \theta$ is
 (1) 0 (2) $1/\sqrt{2}$
 (3) $1/2$ (4) 1
11. If $\sec \theta + \tan \theta = \sqrt{3}$, then $\tan 3\theta$ is
 (1) not defined (2) $1/\sqrt{3}$
 (3) $1/\sqrt{2}$ (4) $\sqrt{3}$
12. If $\sin(60^\circ - \theta) = \cos(\psi - 30^\circ)$, then the value of $\tan(\psi - \theta)$ is (assume that θ and ψ are both positive acute angles with $\psi < 60^\circ$ and $\psi > 30^\circ$).
 (1) $1/\sqrt{3}$ (2) 0
 (3) $\sqrt{3}$ (4) 1
13. If $a \sin \theta + b \cos \theta = c$ then the value of $a \cos \theta - b \sin \theta$ is :
 (1) $\pm \sqrt{-a^2 + b^2 + c^2}$ (2) $\pm \sqrt{a^2 + b^2 - c^2}$
 (3) $\pm \sqrt{a^2 - b^2 - c^2}$ (4) $\pm \sqrt{a^2 - b^2 + c^2}$
14. If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$ where $A > B > 0$ and $A + B$ is an acute angle, then the value B is
 (1) $\frac{\pi}{6}$ (2) $\frac{\pi}{12}$
 (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{2}$
15. $0 \leq \theta \leq 90^\circ$ and $\cos \theta + \sqrt{3} \sin \theta = 2$, then what is the value of θ ?
 (1) $\frac{\pi}{3}$ (2) $\frac{\pi}{4}$
 (3) $\frac{\pi}{6}$ (4) $\frac{\pi}{2}$
16. If $\tan \alpha = n \tan \beta$ and $\sin \alpha = m \sin \beta$, then $\cos^2 \alpha$ is
 (1) $\frac{m^2}{n^2 + 1}$ (2) $\frac{m^2}{n^2}$

- (3) $\frac{m^2-1}{n^2-1}$ (4) $\frac{m^2+1}{n^2+1}$
17. The equation $\cos^2 \theta = \frac{(x+y)^2}{4xy}$ is only possible when
 (1) $x = -y$ (2) $x > y$
 (3) $x = y$ (4) $x < y$
18. If $\sin \alpha + \cos \beta = 2$ ($0^\circ \leq \beta \leq \alpha \leq 90^\circ$), then $\sin\left(\frac{2\alpha+\beta}{3}\right) =$
 (1) $\sin \frac{\alpha}{2}$ (2) $\cos \frac{\alpha}{3}$
 (3) $\sin \frac{\alpha}{3}$ (4) $\cos \frac{2\alpha}{3}$
19. If $\sin \alpha \sec(30^\circ + \alpha) = 1$ ($0 < \alpha < 60^\circ$), then the value of $\sin \alpha + \cos 2\alpha$ is
 (1) 1 (2) $\frac{2+\sqrt{3}}{2\sqrt{3}}$
 (3) 0 (4) $\sqrt{2}$
20. If $\tan \theta = 1$, then the value of $\frac{8\sin \theta + 5\cos \theta}{\sin^3 \theta - 2\cos^3 \theta + 7\cos \theta}$ is
 (1) 2 (2) $2\frac{1}{2}$
 (3) 3 (4) 0.8
21. What is the value of $\sin 390^\circ$
 (1) $1/2$ (2) $-\sqrt{3}/2$
 (3) $-1/2$ (4) $\sqrt{3}/2$
22. What is the value of $\cot 585^\circ$
- (1) 1 (2) -1
 (3) $\sqrt{3}$ (4) $-\sqrt{3}$
23. What is the value of $\cos 1110^\circ$
 (1) $1/2$ (2) $-\sqrt{3}/2$
 (3) $-1/2$ (4) $\sqrt{3}/2$
24. What is the value of $\sin 930^\circ$
 (1) $1/2$ (2) $-\sqrt{3}/2$
 (3) $-1/2$ (4) $\sqrt{3}/2$
25. What is the value of $\sec 330^\circ$
 (1) 2 (2) -2
 (3) $2/\sqrt{3}$ (4) $-2/\sqrt{3}$
26. What is the value of $\cos 660^\circ$
 (1) $1/2$ (2) $\sqrt{3}/2$
 (3) $-1/2$ (4) $\sqrt{3}/2$
27. What is the value of $\sin(-60^\circ)$
 (1) $1/2$ (2) $-\sqrt{3}/2$
 (3) $-1/2$ (4) $\sqrt{3}/2$
28. What is the value of $\cos(-390^\circ)$
 (1) $1/2$ (2) $-\sqrt{3}/2$
 (3) $-1/2$ (4) $\sqrt{3}/2$
29. What is the value of $\tan(-450^\circ)$
 (1) 1 (2) -1
 (3) 0 (4) not defined
30. What is the value of $\sec(-225^\circ)$
 (1) 2 (2) -2
 (3) $\sqrt{2}$ (4) $-\sqrt{2}$

ANSWER KEY

1.	(1)	2.	(1)	3.	(4)	4.	(2)	5.	(4)	6.	(3)	7.	(3)	8.	(2)	9.	(1)	10.	(3)
11.	(1)	12.	(3)	13.	(2)	14.	(2)	15.	(1)	16.	(3)	17.	(3)	18.	(2)	19.	(1)	20.	(1)
21.	(1)	22.	(1)	23.	(4)	24.	(3)	25.	(3)	26.	(1)	27.	(2)	28.	(4)	29.	(4)	30.	(4)

PRACTICE SHEET-5

1. If $\sin A = \frac{1}{\sqrt{5}}$, $\cos B = \frac{3}{\sqrt{10}}$; where A, B being positive acute angles, then what is (A+B) equal to:
 - (1) $\frac{\pi}{6}$
 - (2) $\frac{\pi}{4}$
 - (3) $\frac{\pi}{3}$
 - (4) $\frac{\pi}{2}$
2. What is the value of $\frac{\cos 10^\circ - \sin 10^\circ}{\cos 10^\circ + \sin 10^\circ}$?
 - (1) $\tan 35^\circ$
 - (2) $\tan 10^\circ$
 - (3) $\frac{1}{\sqrt{2}}$
 - (d) 1
3. What is the value of $\sin\left(\frac{5\pi}{12}\right)$?
 - (1) $\frac{\sqrt{3}+1}{2\sqrt{2}}$
 - (2) $\frac{\sqrt{6}+\sqrt{2}}{4}$
 - (3) $\frac{\sqrt{3}+\sqrt{2}}{4}$
 - (4) $\frac{\sqrt{6}+1}{2}$
4. If $\cot \alpha$ and $\cot \beta$ are the roots of equation $x^2 - 3x + 2 = 0$, then what is $\cot(\alpha + \beta)$ equal to?
 - (1) $1/2$
 - (2) $1/3$
 - (3) 2
 - (4) 3
5. What is $\sin(\alpha + \beta) - 2\sin\alpha\cos\beta + \sin(\alpha - \beta) = ?$
 - (1) 0
 - (2) $2\sin\alpha$
 - (3) $2\sin\beta$
 - (4) $\sin\alpha + \sin\beta$
6. If $2\tan A = 3\tan B = 1$, then what is $\tan(A-B)$ equal to?
 - (1) $1/5$
 - (2) $1/6$
 - (3) $1/7$
 - (4) $1/9$
7. What is the value of $\frac{\sin 34^\circ \cos 236^\circ - \sin 56^\circ \sin 124^\circ}{\cos 28^\circ \cos 88^\circ + \cos 178^\circ \sin 208^\circ}$?
 - (1) -2
 - (2) -1
 - (3) 2
 - (4) 1
8. $\tan 54^\circ$ can be expressed as
 - (1) $\frac{\sin 9^\circ + \cos 9^\circ}{\sin 9^\circ - \cos 9^\circ}$
 - (2) $\frac{\sin 9^\circ - \cos 9^\circ}{\sin 9^\circ + \cos 9^\circ}$
 - (3) $\frac{\cos 9^\circ + \sin 9^\circ}{\cos 9^\circ - \sin 9^\circ}$
 - (4) $\frac{\sin 36^\circ}{\cos 36^\circ}$
9. What is the value of $\tan 75^\circ + \cot 75^\circ$?
 - (1) 2
 - (2) 4
 - (3) $2\sqrt{3}$
 - (4) $4\sqrt{3}$
10. What is $\frac{2\tan\theta}{1+\tan^2\theta}$ equal to?
 - (1) $\cos 2\theta$
 - (2) $\tan 2\theta$
 - (3) $\sin 2\theta$
 - (4) $\operatorname{cosec} 2\theta$
11. If $\frac{\sin(x+y)}{\sin(x-y)} = \frac{a+b}{a-b}$, then what is $\frac{\tan x}{\tan y}$ equal to?
 - (1) $\frac{a}{b}$
 - (2) $\frac{b}{a}$
 - (3) $\frac{a+b}{a-b}$
 - (4) $\frac{a-b}{a+b}$
12. What is $\sin 105^\circ + \cos 105^\circ$ equal to?
 - (1) $\sin 50^\circ$
 - (2) $\cos 50^\circ$
 - (3) $1/\sqrt{2}$
 - (4) 0
13. What is $\frac{\sin 5x - \sin 3x}{\cos 5x + \cos 3x}$ equal to?
 - (1) $\sin x$
 - (2) $\cos x$
 - (3) $\tan x$
 - (4) $\cot x$
14. If $\sin x = \frac{1}{\sqrt{5}}$, $\sin y = \frac{1}{\sqrt{10}}$, where $0 < x < \frac{\pi}{2}$, $0 < y < \frac{\pi}{2}$, then what is $(x+y)$ equal to?
 - (1) π
 - (2) $\pi/2$
 - (3) $\pi/4$
 - (4) 0
15. If $\tan(\alpha + \beta) = 2$ and $\tan(\alpha - \beta) = 1$, then $\tan(2\alpha)$ is equal to:
 - (1) -3
 - (2) -2
 - (3) $-\frac{1}{3}$
 - (d) 1
16. What is $\frac{1 - \tan 2^\circ \cot 62^\circ}{\tan 152^\circ - \cot 88^\circ}$ equal to?
 - (1) $\sqrt{3}$
 - (2) $-\sqrt{3}$
 - (3) $\sqrt{2}-1$
 - (4) $1 - \sqrt{2}$

17. Consider the following statements:
 1.If ABC is an equilateral triangle, then $3\tan(A+B)\tan C = 1$
 2.If ABC is a triangle in which $A = 78^\circ$, $B = 66^\circ$, then $\tan\left(\frac{A+C}{2}\right) < \tan A$
 3.If ABC is any triangle, then $\tan\left(\frac{A+B}{2}\right)\sin\left(\frac{C}{2}\right) < \cos\left(\frac{C}{2}\right)$
 which of the above statements is/are correct?
 (1) Only 1 (2) Only 2
 (3) 1 and 2 (4) 2 and 3
18. What is $\frac{\cos 7x - \cos 3x}{\sin 7x - 2\sin 5x + \sin 3x}$ equal to?
 (1) $\tan x$ (2) $\cot x$
 (3) $\tan 2x$ (4) $\cot 2x$
19. If $A + B + C = \pi$, then what is $\cos(A+B) + \cos C$ equal to?
 (1) 0 (2) $2\cos C$
 (3) $\cos C - \sin C$ (4) $2\sin C$
20. What is $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ$ equal to?
 (1) 2 (2) 1
 (3) $\frac{1}{2}$ (4) 0
21. What is $\sin^2(3\pi) + \cos^2(4\pi) + \tan^2(5\pi)$ equal to?
 (1) 0 (2) 1
 (3) 2 (4) 3
22. Consider the following statements:
 I. Value of $\sin\theta$ oscillates between -1 and 1
 II. Value of $\cos\theta$ oscillates between 0 to 1
 Which of the above statements is/are correct?
 (1) Only I (2) Only II
 (3) Both I and II (4) Neither I nor II
23. What is $(1 - \sin^2 \theta)(1 + \tan^2 \theta)$ equal to?
 (1) $\sin^2 \theta$ (2) $\cos^2 \theta$
 (3) $\tan^2 \theta$ (4) 1
24. What is $\sin^2 20^\circ + \sin^2 70^\circ$ equal to?
 (1) 1 (2) 0
 (3) -1 (4) $\frac{1}{2}$
25. If $\cos x = 1/3$. Then, what is $\sin x \cdot \cot x \cdot \operatorname{cosec} x \cdot \tan x$ equal to?
 (1) $2/3$ (2) $3/2$
 (c) 2 (4) 1
26. What is $\frac{\cot 54^\circ}{\tan 36^\circ} + \frac{\tan 20^\circ}{\cot 70^\circ}$ equal to?
 (1) 0 (2) 1
 (3) 2 (4) 3
27. What is $\tan 15^\circ$ equal to?
 (1) $2 - \sqrt{3}$ (2) $2 + \sqrt{3}$
 (3) $1 - \sqrt{3}$ (4) $1 + \sqrt{3}$
28. What is the value of $\tan 105^\circ$?
 (1) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$ (2) $\frac{\sqrt{3}+1}{1-\sqrt{3}}$
 (3) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$ (4) $\frac{\sqrt{3}+2}{\sqrt{3}-1}$
29. If $\sin\theta + 2\cos\theta = 1$, then what is $2\sin\theta - \cos\theta$ equal to?
 (1) 0 (2) 1
 (3) 2 (4) 4
30. What is $\frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$ equal to?
 (1) $\sin x \cos x$ (2) $\tan x$
 (3) $\sin x$ (4) $\cos x$

ANSWER KEY

1.	(2)	2.	(1)	3.	(1)	4.	(2)	5.	(1)	6.	(3)	7.	(1)	8.	(3)	9.	(2)	10.	(3)
11.	(1)	12.	(3)	13.	(3)	14.	(3)	15.	(1)	16.	(2)	17.	(2)	18.	(2)	19.	(1)	20.	(4)
21.	(2)	22.	(1)	23.	(4)	24.	(1)	25.	(4)	26.	(3)	27.	(1)	28.	(2)	29.	(3)	30.	(4)

PRACTICE SHEET-6

1. The minimum value of $2 \sin^2 \theta + 3 \cos^2 \theta$ is
 (1) 0 (2) 3
 (3) 2 (4) 1
2. If $\cos x + \cos y = 2$, the value of $\sin x + \sin y$ is
 (1) 0 (2) 1
 (3) 2 (4) -1
3. If $0^\circ < \theta < 90^\circ$, the value of $\sin \theta + \cos \theta$ is
 (1) equal to 1 (2) greater than 1
 (3) less than 1 (4) equal to 2
4. Maximum value of $(2 \sin \theta + 3 \cos \theta)$ is
 (1) 2 (2) $\sqrt{13}$
 (3) $\sqrt{15}$ (4) 1
5. The minimum value of $4 \tan^2 \theta + 9 \cot^2 \theta$ is equal to
 (1) 0 (2) 5
 (3) 12 (4) 13
6. If θ be acute and $\tan \theta + \cot \theta = 2$, then the value of $\tan^5 \theta + \cot^5 \theta$ is
 (1) 1 (2) 2
 (3) 3 (4) 4
7. If $\sin \theta + \operatorname{cosec} \theta = 2$, then value of $\sin^{100} \theta + \operatorname{cosec}^{100} \theta$ is equal to :
 (1) 1 (2) 2
 (3) 3 (4) 100
8. What is the minimum value of $8 \sin^2 \theta + 18 \operatorname{cosec}^2 \theta$
 (1) 18 (2) 24
 (3) 3 (4) 100
9. If $A = \sin^2 \theta + \cos^4 \theta$, for any value of θ , then the value of A is
 (1) $1 \leq A \leq 2$ (2) $3/4 \leq A \leq 1$
 (3) $13/16 \leq A \leq 1$ (4) $3/4 \leq A \leq 13/16$
10. The greatest value of $\sin^4 \theta + \cos^4 \theta$ is
 (1) $\sqrt{2}$ (2) 3
 (3) $1/2$ (4) 1
11. If $\sec \theta = x + \frac{1}{4x}$ ($0^\circ < \theta < 90^\circ$) then $\sec \theta + \tan \theta$ is equal to
 (1) $\frac{x}{2}$ (2) $2x$
- (3) x (4) $\frac{1}{2x}$
12. $\sin^2 \theta - 3 \sin \theta + 2 = 0$ will be true if
 (1) $0 \leq \theta \leq 90$ (2) $0 < \theta < 90$
 (3) $\theta = 0^\circ$ (4) $\theta = 90^\circ$
13. If $\frac{\pi}{4} < \theta < \frac{\pi}{2}$, then what is $\sqrt{1 - 2 \sin \theta \cos \theta}$ equal to?
 (a) $\cos \theta - \sin \theta$ (b) $\sin \theta - \cos \theta$
 (c) $\pm(\cos \theta - \sin \theta)$ (d) $\cos \theta \sin \theta$
14. If $0 < x < \pi/2$, then $(\cos x + \sec x)$ is:
 (a) > 2 (b) < 2
 (c) ≥ 2 (d) ≤ 2
15. If $\tan^2 \alpha = 1 + 2 \tan^2 \beta$, then $\sqrt{2} \cos \alpha - \cos \beta$ is equal to
 (1) 0 (2) $\sqrt{2}$
 (3) 1 (4) -1
16. If $0 < x < \frac{\pi}{2}$ and $\sec x = \operatorname{cosec} y$, then the value of $\sin(x + y)$ is
 (1) 0 (2) 1
 (3) $1/2$ (4) $1/\sqrt{3}$
17. If $\cos^4 \theta - \sin^4 \theta = \frac{2}{3}$, then the value of $2 \cos^2 \theta - 1$ is
 (1) 0 (2) 1
 (3) $2/3$ (4) $3/2$
18. If $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = 3$, then the value of $\sin^4 \theta - \cos^4 \theta$ is
 (a) $1/5$ (2) $2/5$
 (3) $3/5$ (4) $4/5$
19. If $2 \sin\left(\frac{\pi x}{2}\right) = x^2 + \frac{1}{x^2}$, then the value of $\left(x - \frac{1}{x}\right)$ is
 (1) -1 (2) 2
 (3) 1 (4) 0

20. Which one of the following is true for $0^\circ < \theta < 90^\circ$?
 (1) $\cos\theta \leq \cos^2\theta$ (2) $\cos\theta > \cos^2\theta$
 (3) $\cos\theta < \cos^2\theta$ (4) $\cos\theta \geq \cos^2\theta$
21. If $(\sin\alpha + \operatorname{cosec}\alpha)^2 + (\cos\alpha + \sec\alpha)^2 = k + \tan^2\alpha + \cot^2\alpha$, then the value of k is
 (1) 1 (2) 7
 (3) 3 (4) 5
22. If $0^\circ < \theta < 90^\circ$ and $\sec\theta = 3 \operatorname{cosec}^2\theta$, then θ is
 (1) $\frac{\pi}{6}$ (2) $\frac{\pi}{4}$
 (3) $\frac{\pi}{3}$ (4) $\frac{\pi}{5}$
23. $\sqrt{\frac{1+\sin\theta}{1-\sin\theta}} + \sqrt{\frac{1-\sin\theta}{1+\sin\theta}}$ is equal to
 (1) $2 \cos\theta$ (2) $2 \sin\theta$
 (3) $2 \cot\theta$ (4) $2 \sec\theta$
24. If $0^\circ < \theta < 90^\circ$ then the value of $\tan^2\theta + \cot^2\theta - \sec^2\theta \operatorname{cosec}^2\theta$ is
 (1) 0 (2) 1
 (3) 2 (4) -2
25. If α and β are positive acute angles, $\sin(4\alpha - \beta) = 1$ and $\cos(2\alpha + \beta) = 1/2$, then the value of $\sin(\alpha + 2\beta)$ is
 (1) 0 (2) 1
 (3) $\sqrt{3}/2$ (4) $1/\sqrt{2}$
26. The minimum value of $\cos^2 x + \cos^2 y - \cos^2 z$ is:
 (a) -1 (b) 0
 (c) 1 (d) 2
27. If $r \sin\theta = \frac{7}{2}$ and $r \cos\theta = \frac{7\sqrt{3}}{2}$, then the value of r is
 (1) 4 (2) 3
 (3) 5 (4) 7
28. If $0 \leq \theta \leq 90^\circ$ and $4 \cos^2\theta - 4\sqrt{3}\cos\theta + 3 = 0$ then the value of θ is
 (1) 30° (2) 45°
 (3) 90° (4) 60°
29. If $\sec\theta - \cos\theta = 3/2$ where θ is a positive acute angle, then the value of $\sec\theta$ is
 (1) $-1/2$ (2) 1
 (3) 2 (4) 0
30. If $\tan(5x-10^\circ) = \cot(5y+20^\circ)$, then the value of $(x+y)$ is
 (1) 15° (2) 16°
 (3) 24° (4) 20°

ANSWER KEY

1.	(3)	2.	(1)	3.	(2)	4.	(2)	5.	(3)	6.	(2)	7.	(2)	8.	(2)	9.	(2)	10.	(4)
11.	(2)	12.	(4)	13.	(2)	14.	(3)	15.	(1)	16.	(2)	17.	(3)	18.	(3)	19.	(4)	20.	(2)
21.	(2)	22.	(3)	23.	(4)	24.	(4)	25.	(4)	26.	(1)	27.	(4)	28.	(1)	29.	(3)	30.	(2)

CDS PYQ

1. If $\cos A + \cos^2 A = 1$, then what is the value of $2(\sin^2 A + \sin^4 A)$?

[CDS-2013-1]

- (a) 4 (b) 2
(c) 1 (d) $\frac{1}{2}$

2. $(1-\tan A)^2 + (1+\tan A)^2 + (1-\cot A)^2 + (1+\cot A)^2$ is equal to:

[CDS-2013-1]

- (a) $\sin^2 A \cos^2 A$ (b) $\sec^2 A \operatorname{cosec}^2 A$
(c) $2 \sec^2 A \operatorname{cosec}^2 A$ (d) None of these

Directions (for next four): Read the following information carefully to answer the questions that follow.

The angles A, B, C and D of a quadrilateral ABCD are in the ratio 1:2:4:5.

A=30°, B=60°, C=120°, D=150°

3. What is the value of $\cos(A + B)$?

[CDS-2013-1]

- (a) 0 (b) $\frac{1}{2}$
(c) 1 (d) None of these

4. What is the value of $\operatorname{cosec}(C - D + B)$?

[CDS-2013-1]

- (a) 1 (b) 2
(c) 3 (d) 4

5. Consider the following statements:

I. ABCD is a cyclic quadrilateral

II. $\sin(B - A) = \cos(D - C)$

Which of the above statement is/are correct?

[CDS-2013-1]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

6. What is the value of $\sec^2 D - \tan^2 D$?

[CDS-2013-1]

- (a) $\frac{1}{2}$ (b) $\frac{2}{3}$
(c) 1 (d) None of these

7. What is the value of $\frac{\tan A - \sin A}{\sin^3 A}$?

[CDS-2013-1]

- (a) $\frac{\sec A}{1 - \cos A}$ (b) $\frac{\sec A}{1 + \cos^2 A}$
(c) $\frac{\sec A}{1 + \cos A}$ (d) None of these

8. Consider the following $\sin \theta \leq \theta \leq 90^\circ$.

I. The value of $\sin \theta + \cos \theta$ is always greater than 1

II. The value of $\tan \theta + \cot \theta$ is always greater than 1

Which of the above statement is/are correct?

[CDS-2013-1]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

9. If $\sin A = \frac{3}{5}$ and A is an acute angle, then $\tan A + \sec A$ is equal to:

[CDS-2013-1]

- (a) 0 (b) 1
(c) 2 (d) -1

10. If $\sin \theta = \frac{x^2 - y^2}{x^2 + y^2}$, then which one of the following is:

[CDS-2013-1]

- (a) $\cos \theta = \frac{2xy}{x^2 - y^2}$ (b) $\cos \theta = \frac{2xy}{x^2 + y^2}$
(c) $\cos \theta = \frac{x - y}{x^2 + y^2}$ (d) $\cos \theta = \frac{xy(x - y)}{x^2 + y^2}$

11. If $a^2 = \frac{1 + 2\sin \theta \cos \theta}{1 - 2\sin \theta \cos \theta}$, then what is the value of

$$\frac{a+1}{a-1}?$$

[CDS-2013-1]

- (a) $\sec \theta$ (b) 1
(c) 0 (d) $\tan \theta$

12. If $5 \sin \theta + 12 \cos \theta = 13$, then what is $5 \cos \theta - 12 \sin \theta$ equal to?

[CDS-2013-2]

- (a) -2 (b) -1
(c) 0 (d) 1

13. If $4 \tan \theta = 3$, then what is $\frac{4 \sin \theta - \cos \theta}{4 \sin \theta + 9 \cos \theta}$ equal to?

[CDS-2013-2]

- (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{1}{4}$ (d) $\frac{1}{6}$

14. If $\sin \theta - \cos \theta = 0$, then what is $\sin^4 \theta + \cos^4 \theta$ equal to?

[CDS-2013-2]

- (a) 1 (b) $\frac{3}{4}$
(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

15. Consider the following statements:

I. $\tan \theta$ increases faster than $\sin \theta$ as θ increases

II. The value of $\sin \theta + \cos \theta$ is always greater than I

Which of the above statement is/are correct?

[CDS-2013-2]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

16. What is $\frac{(\sin \theta + \cos \theta)(\tan \theta + \cot \theta)}{\sec \theta + \operatorname{cosec} \theta}$ equal to?

[CDS-2013-2]

- (a) 1 (b) 2
(c) $\sin \theta$ (d) $\cos \theta$

17. What is $\frac{\cos^2(45^\circ + \theta) + \cos^2(45^\circ - \theta)}{\tan(60^\circ + \theta) \tan(30^\circ - \theta)}$ equal to?

[CDS-2013-2]

- (a) -1 (b) 0
(c) 1 (d) 2

18. What is $\sin^6 \theta + \cos^6 \theta + 3 \sin^2 \theta \cos^2 \theta$ equal to?

[CDS-2013-2]

- (a) 0 (b) 1
(c) 2 (d) 4

19. What is $\frac{(1 + \sec \theta - \tan \theta) \cos \theta}{(1 + \sec \theta + \tan \theta)(1 - \sin \theta)}$ equal to?

[CDS-2013-2]

- (a) 1 (b) 2
(c) $\tan \theta$ (d) $\cot \theta$

20. If $\sin \theta + \cos \theta = \sqrt{3}$, then what is $\tan \theta + \cot \theta$ equal to?

[CDS-2013-2]

- (a) 1 (b) $\sqrt{2}$
(c) 2 (d) $\sqrt{3}$

21. If $\tan \theta + \sec \theta = m$, then what is $\sec \theta$ equal to?

[CDS-2013-2]

- (a) $\frac{m^2 - 1}{2m}$ (b) $\frac{m^2 + 1}{2m}$
(c) $\frac{m + 1}{m}$ (d) $\frac{m^2 + 1}{m}$

22. What is $\operatorname{cosec}(75^\circ + \theta) - \sec(15^\circ - \theta)$ equal to?

[CDS-2013-2]

- (a) 0 (b) 1
(c) $2 \sin \theta$ (d) $2 \cos \theta$

23. If $\triangle ABC$ is right angled at C, then what is $\cos(A + B) + \sin(A + B)$ equal to?

[CDS-2013-2]

- (a) 0 (b) $\frac{1}{2}$
(c) 1 (d) 2

24. If α , β and γ are acute angles such that $\sin \alpha = \frac{\sqrt{3}}{2}$, $\cos \beta = \frac{\sqrt{3}}{2}$ and $\tan \gamma = 1$, then what is $\alpha + \beta + \gamma$ equal to?

[CDS-2013-2]

- (a) 105° (b) 120°
(c) 135° (d) 150°

25. The value of $\cos 25^\circ - \sin 25^\circ$ is:

[CDS-2014-1]

- (a) Positive but less than I
(b) Positive but greater than I

- (c) Negative
(d) 0

26. In a right angled $\triangle ABC$, right angle at B, if $\cos A = \frac{4}{5}$, then what is the value of $\sin C$?

[CDS-2014-1]

- (a) $\frac{3}{5}$ (b) $\frac{4}{5}$
(c) $\frac{3}{5}$ (d) $\frac{2}{5}$

27. If α and β are complementary angle, then what is $\sqrt{\cos \alpha \operatorname{cosec} \beta - \cos \alpha \sin \beta}$ equal to?

[CDS-2014-1]

- (a) $\sec \beta$ (b) $\cos \alpha$
(c) $\sin \alpha$ (d) $-\tan \beta$

28. If $2\cot \theta = 3$, then what is $\frac{2\cos \theta - \sin \theta}{2\cos \theta + \sin \theta}$ equal to?

[CDS-2014-1]

- (a) $\frac{2}{3}$ (b) $\frac{1}{3}$
(c) $\frac{1}{2}$ (d) $\frac{3}{4}$

29. If $\sin \theta \cos \theta = \frac{1}{2}$, then what is $\sin^6 \theta + \cos^6 \theta$ equal to?

[CDS-2014-1]

- (a) 1 (b) 2
(c) 3 (d) $\frac{1}{4}$

30. If $\sec \theta + \tan \theta = 2$, then what is the value of $\sec \theta$?

[CDS-2014-1]

- (a) $\frac{3}{2}$ (b) $\sqrt{2}$
(c) $\frac{5}{\sqrt{2}}$ (d) $\frac{5}{4}$

31. What is $\operatorname{cosec} (75^\circ + \theta) - \sec (15^\circ - \theta) - \tan (55^\circ + \theta) + \cot (35^\circ - \theta)$ equal to?

[CDS-2014-1]

- (a) -1 (b) 0

- (c) 1 (d) $\frac{3}{2}$

32. If $\sin \theta + 2\cos \theta = 1$, where $0 < \theta < \frac{\pi}{2}$, what is $2\sin \theta - \cos \theta$ equal to?

[CDS-2014-1]

- (a) -1 (b) $\frac{1}{2}$
(c) 2 (d) 1

33. If $\cos x + \sec x = 2$, then what $\cos^n x + \sec^n x$ equal to, where n is a positive integer?

[CDS-2014-1]

- (a) 2 (b) 2^{n-2}
(c) 2^{n-1} (d) 2^n

34. What is $\sin 25^\circ \sin 35^\circ \sec 65^\circ \sec 55^\circ$ equal to?

[CDS-2014-1]

- (a) -1 (b) 0
(c) $\frac{1}{2}$ (d) 1

35. If $\tan 8\theta = \cot 2\theta$, where $\theta < 8\theta < \frac{\pi}{2}$, then what is the value of $\tan 5\theta$?

[CDS-2014-1]

- (a) $\frac{1}{\sqrt{3}}$ (b) 1
(c) $\sqrt{3}$ (d) 0

36. If $\sin (A + B) = 1$ where, $0 < B < 45^\circ$, then what is $\cos (A - B)$ equal to?

[CDS-2014-1]

- (a) $\sin 2B$ (b) $\sin B$
(c) $\cos 2B$ (d) $\cos B$

37. At what point of time after 3 O'clock, hour hand and the minute hand of a clock occur at right angles for the first time?

[CDS-2014-1]

- (a) 9 O'clock (b) $4\text{h } 37\frac{1}{6}\text{ min}$
(c) $3\text{h } 30\frac{8}{11}\text{ min}$ (d) $3\text{h } 32\frac{8}{11}\text{ min}$

38. If $0 < \theta < \frac{\pi}{4}$, then what is $\sqrt{1-2\sin\theta\cos\theta}$ equal to?

[CDS-2014-2]

- (a) $\cos\theta - \sin\theta$ (b) $\sin\theta - \cos\theta$
(c) $\pm(\cos\theta - \sin\theta)$ (d) $\cos\theta \sin\theta$

39. If $\tan\theta + \cot\theta = 2$, then what is $\sin\theta + \cos\theta$ equal to:

[CDS-2014-2]

- (a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{3}}$
(c) $\sqrt{2}$ (d) 1

40. What is $\frac{\sec x}{\cot x + \tan x}$ equal to?

[CDS-2014-2]

- (a) $\sin x$ (b) $\cos x$
(c) $\tan x$ (d) $\cot x$

41. What is $\frac{\sin x - \cos x + 1}{\sin x + \cos x - 1}$ equal to?

[CDS-2014-2]

- (a) $\frac{\sin x - 1}{\cos x}$ (b) $\frac{\sin x + 1}{\cos x}$
(c) $\frac{\sin x - 1}{\cos x + 1}$ (d) $\frac{\sin x + 1}{\cos x + 1}$

42. What is $(\sin^2 x - \cos^2 x)(1 - \sin^2 x \cos^2 x)$ equal to?

[CDS-2014-2]

- (a) $\sin^4 x - \cos^4 x$ (b) $\sin^6 x - \cos^6 x$
(c) $\cos^8 x - \sin^8 x$ (d) $\sin^8 x - \cos^8 x$

43. What is $(\sin x \cos y + \cos x \sin y)(\sin x \cos y - \cos x \sin y)$ equal to?

[CDS-2014-2]

- (a) $\cos^2 x - \cos^2 y$ (b) $\cos^2 x - \sin^2 y$
(c) $\sin^2 x - \cos^2 y$ (d) $\sin^2 x - \sin^2 y$

44. What is $(1 + \cot x - \operatorname{cosec} x)(1 + \tan x + \sec x)$ equal to?

[CDS-2014-2]

- (a) 1 (b) 2
(c) $\sin x$ (d) $\cos x$

45. What is $(\operatorname{cosec} x - \sin x)(\sec x - \cos x)(\tan x + \cot x)$ equal to?

[CDS-2014-2]

- (a) $\sin x + \cos x$ (b) $\sin x - \cos x$
(c) 2 (d) 1

46. Consider the following statements:

I. $\sin 1^\circ > \sin 1$

II. $\cos 1^\circ < \cos 1$

Which of the above statements is /are correct?

[CDS-2014-2]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

47. If $\sin x + \operatorname{cosec} x = 2$, then what is $\sin^9 x + \operatorname{cosec}^9 x$ equal to?

[CDS-2014-2]

- (a) 2 (b) 18
(c) 512 (d) 1024

48. If $\sin x + \cos x = p$ and $\sin^3 x + \cos^3 x = q$, then what is $p^3 - 3p$ equal to?

[CDS-2014-2]

- (a) 0 (b) $-2q$
(c) $2q$ (d) $4q$

49. If $\tan A + \cot A = 4$, then $\tan^4 A + \cot^4 A$ equal to:

[CDS-2014-2]

- (a) 110 (b) 191
(c) 80 (d) 194

50. If $\tan(A + B) = \sqrt{3}$ and $\tan A = 1$, then $\tan(A - B)$ is equal to:

[CDS-2015-1]

- (a) 0 (b) 1
(c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{2}$

51. If $\cos A = \tan B$, $\cos B = \tan C$ and $\cos C = \tan A$ then $\sin A$ is equal to:

[CDS-2015-1]

- (a) $\frac{\sqrt{5}-1}{4}$ (b) $\frac{\sqrt{5}-1}{2}$
(c) $\frac{\sqrt{3}-1}{4}$ (d) none

52. If $\frac{3 - \tan^2 A}{1 - 3\tan^2 A} = K$ where K is a real number, then $\operatorname{cosec} A(3 \sin A - 4 \sin^3 A)$ is equal to:

[CDS-2015-1]

- (a) $\frac{2K}{K-1}$
(b) $\frac{2K}{K-1}$, where $\frac{1}{3} \leq K \leq 3$
(c) $\frac{2K}{K-1}$, where $K < \frac{1}{3}$ or $K > 3$

(d) $\frac{2K}{K+1}$

53. If $P = \sqrt{\frac{1-\sin x}{1+\sin x}}$, $q = \frac{1-\sin x}{\cos x}$, $r = \frac{\cos x}{1+\sin x}$, then

which of the following is/are correct?

1. $p = q = r$

2. $p^2 = qr$

Select the correct answer using the code given below.

[CDS-2015-1]

(a) Only 1

(b) Only 2

(c) Both 1 and 2

(d) Neither 1 nor 2

54. Consider the following:

1. $\frac{\cos A}{1-\tan A} + \frac{\sin A}{1-\cot A} = \cos A + \sin A$.

2. $(1-\sin A - \cos A)^2 = 2(1-\sin A)(1+\cos A)$
Which of the above is/are identify / identities ?

[CDS-2015-1]

(a) 1 Only

(b) 2 Only

(c) Both 1 and 2

(d) Neither 1 nor 2

55. ABC is a triangle right angled at B and AB : BC = 3 : 4. What is $\sin A + \sin B + \sin C$ equal to?

[CDS-2015-1]

(a) 2

(b) $\frac{11}{5}$

(c) $\frac{12}{5}$

(d) 3

56. The value of $\operatorname{cosec}^2 67^\circ + \sec^2 57^\circ - \cot^2 33^\circ - \tan^2 23^\circ$ is:

[CDS-2015-1]

(a) $2\sqrt{2}$

(b) 2

(c) $\sqrt{2}$

(d) 0

57. Consider the following statements:

1. There exists at least one value of x between 0 and $\pi/2$ which satisfies the equation $\sin^4 x - 2\sin^2 x - 1 = 0$.

2. $\sin 1.5$ is greater than $\cos 1.5$. Which of the above statement is/are correct?

[CDS-2015-1]

(a) 1 Only

(b) 2 Only

(c) Both 1 and 2

(d) Neither 1 nor 2

58. If $\sin x + \cos x = c$ then $\sin^6 x + \cos^6 x$ is equal to:

[CDS-2015-1]

(a) $\frac{1+6c^2-3c^4}{16}$

(b) $\frac{1+6c^2-3c^4}{4}$

(c) $\frac{1+6c^2+3c^4}{16}$

(d) $\frac{1+6c^2+3c^4}{4}$

59. Consider the following statements:

1. There exists no value of x such that

$$\frac{1}{1-\sin x} = 4 + 2\sqrt{3}, 0 < x < \frac{\pi}{2}$$

2. $\sin x = 3^{\sin^2 x}$ does not hold good for any real x . Which of the above statement is/are correct?

[CDS-2015-1]

(a) 1 Only

(b) 2 Only

(c) Both 1 and 2

(d) Neither 1 nor 2

60. The complement angle of 80° is:

[CDS-2015-1]

(a) $\frac{\pi}{9}$

(b) $\frac{5\pi}{9}$

(c) $\frac{\pi}{18}$

(d) $\frac{9}{5\pi}$

61. A triangle ABC is inscribed in a circle. If sum of the squares of sides of the triangle is equal to twice the square of the diameter, then $\sin^2 A + \sin^2 B + \sin^2 C$ is equal to?

[CDS-2015-1]

(a) 2

(b) 3

(c) 4

(d) None of these

62. At 8:30, the hour hand and the minute hand of a clock form an angle of:

[CDS-2015-1]

(a) 80°

(b) 75°

(c) 70°

(d) 60°

63. A person goes to a market between 4 pm and 5 pm. When he comes back, he finds that the hour hand and minute hand have interchanged their position. For how much time (approximately) was he out of his house?

[CDS 2015 (1)]

(a) 55.25 minutes

(b) 55.30 minutes

(c) 55.34 minutes

(d) 55.38 minutes

64. Consider the following statements:

1. $\frac{1+\tan^2 \theta}{1+\cot^2 \theta} = \left(\frac{1-\tan \theta}{1-\cot \theta}\right)^2$ is true for all $0 < \theta < \frac{\pi}{2}$, $\theta \neq \frac{\pi}{4}$

$$\frac{\pi}{2}, \theta \neq \frac{\pi}{4}$$

2. $\cot \theta - \frac{1}{\tan \theta}$ is true for $\theta = 45^\circ$ only

Which of the above statements is/are correct?

[CDS-2015-2]

- (a) 1 Only (b) 1
(c) $\tan^2 \theta$ (d) $\sin^2 \theta$

65. If $x = a \cos \theta$ and $y = b \cot \theta$, then $(ax^{-1} - by^{-1})(ax^{-1} + by^{-1})$ is equal to:

[CDS-2015-2]

- (a) 0 (b) 1
(c) $\tan^2 \theta$ (d) $\sin^2 \theta$

66. $\frac{\cos \theta}{1 - \sin \theta}$ is equal $\left(\text{where } \theta \neq \frac{\pi}{2} \right)$:

[CDS-2015-2]

- (a) $\frac{\tan \theta - 1}{\tan \theta + 1}$ (b) $\frac{1 + \sin \theta}{\cos \theta}$
(c) $\frac{\tan \theta + 1}{\tan \theta - 1}$ (d) $\frac{1 + \cos \theta}{\sin \theta}$

67. If $\tan(x + 40)^\circ \tan(x + 20)^\circ \tan(3x)^\circ \tan(70 - x)^\circ \tan(50 - x)^\circ = 1$, then the value of x is equal to:

[CDS-2015-2]

- (a) 30 (b) 20
(c) 15 (d) 10

68. If θ is an acute angle and $\sin \theta \cos \theta = 2 \cos^3 \theta - 1.5 \cos \theta$, then what is $\sin \theta$ equal to?

[CDS-2015-2]

- (a) $\frac{\sqrt{5} - 1}{4}$ (b) $\frac{1 - \sqrt{5}}{4}$
(c) $\frac{\sqrt{5} + 1}{4}$ (d) $-\frac{\sqrt{5} + 1}{4}$

69. Consider the following statements:

1. $\sin 66^\circ$ is less than $\cos 66^\circ$
2. $\sin 26^\circ$ is less than $\cos 26^\circ$

Which of the above statement is/are correct?

[CDS-2015-2]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

70. If a and b are positive, then the relation $\sin \theta = \frac{2a + 3b}{3b}$ is:

[CDS-2015-2]

- (a) Not possible
(b) Possible only if $a = b$
(c) Possible if $a > b$
(d) Possible if $a < b$

71. If $\tan \theta + \sec \theta = 2$, then $\tan \theta$ is equal to:

[CDS-2015-2]

- (a) $\frac{3}{4}$ (b) $\frac{5}{4}$

- (c) $\frac{3}{2}$ (d) $\frac{5}{2}$

72. A clock is started at noon. By 10 minutes past 5, through what angle, the hour hand moves?

[CDS-2015-2]

- (a) 160° (b) 145°
(c) 150° (d) 150°

73. The minimum value of $\cos^2 x + \cos^2 y - \cos^2 z$ is:

[CDS-2015-2]

- (a) -1 (b) 0
(c) 1 (d) 2

74. The value of:

$32 \cot^2 \left(\frac{\pi}{4} \right) - 8 \sec^2 \left(\frac{\pi}{3} \right) + 8 \cos^3 \left(\frac{\pi}{6} \right)$ is equal to:

[CDS-2015-2]

- (a) $\sqrt{3}$ (b) $2\sqrt{3}$
(c) 3 (d) $3\sqrt{3}$

75. $\left(\frac{\sin 35^\circ}{\cos 55^\circ} \right)^2 - \left(\frac{\cos 55^\circ}{\sin 35^\circ} \right)^2 + 2 \sin 30^\circ$ is equal to:

[CDS-2016-1]

- (a) -1 (b) 0
(c) 1 (d) 2

76. If $\frac{x}{a} - \frac{y}{b} \tan \theta = 1$ and $\frac{x}{a} \tan \theta + \frac{y}{b} = 1$, then the value of $\frac{x^2}{a^2} + \frac{y^2}{b^2}$ is:

[CDS-2016-1]

- (a) $2 \sec^2 \theta$ (b) $\sec^2 \theta$
(c) $2 \cos^2 \theta$ (d) $2 \cos^2 \theta$

77. Consider the following:

1. $\sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} = \operatorname{cosec} \theta - \cot \theta$

2. $\sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}} = \operatorname{cosec} \theta + \cot \theta$

Which of the above is/are identity identities?

[CDS-2016-1]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

78. If $p = \cot \theta + \tan \theta$ and $q = \sec \theta - \cos \theta$, then $(p^2 q)^{2/3} (q^2 p)^{2/3}$ is equal to:

[CDS-2016-1]

- (a) 0 (b) 1
(c) 2 (d) 3

79. If $\frac{\cos^2 \theta - 3\cos \theta + 2}{\sin^2 \theta} = 1$, where $0 < \theta < \frac{\pi}{2}$, then

which of the following statements is/are correct?

1. There are two values of θ satisfying the above equation

2. $\theta = 60^\circ$ is satisfied by the above equation

Select the correct answer using the code given below:

[CDS-2016-1]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

80. Which of the following is correct in respect of the equation $3 - \tan^2 \theta = \alpha (1 - 3\tan^2 \theta)$? (given that α is a real number)

[CDS-2016-1]

(a) $\alpha \in \left[\frac{1}{3}, 3\right]$

(b) $\alpha \in \left[-\infty, \frac{1}{3}\right] \cup [3, \infty]$

(c) $\alpha \in \left(-\infty, \frac{1}{3}\right] \cup [3, \infty)$

(d) None of the above

81. If $\tan \theta + \cot \theta = \frac{4}{\sqrt{3}}$, where $0 < \theta < \frac{\pi}{2}$, then $\sin \theta + \cos \theta$ is equal to:

[CDS-2016-1]

(a) 1 (b) $\frac{\sqrt{3}-1}{2}$

(c) $\frac{\sqrt{3}+1}{2}$ (d) $\sqrt{2}$

82. A person goes to a market between 4pm and 5pm. When he comes back, he finds that the hour hand and the minute hand of the clock have interchanged their positions. For how much time (approximately) was he out of his house?

[CDS 2016 (1)]

- (a) 55.38 minutes (b) 55.48 minutes
(c) 55.57 minutes (d) 55.67 minutes

83. If $\sin \theta + \cos \theta = \frac{\sqrt{7}}{2}$, then what is $\sin \theta - \cos \theta$ equal to:

[CDS-2016-2]

- (a) 0 (b) $\frac{1}{2}$
(c) 1 (d) $\sqrt{2}$

84. If $\sin x + \sin^2 x = 1$, then what is the value of $\cos^8 x + 2\cos^6 x + \cos^4 x$?

[CDS-2016-2]

- (a) 0 (b) 1
(c) 1 (d) 4

85. What is the value of : $\operatorname{cosec}^2 68^\circ + \sec^2 56^\circ - \cot^2 34^\circ - \tan^2 22^\circ$?

[CDS-2016-2]

- (a) 0 (b) $1/2$
(c) 1 (d) 2

86. If $2y \cos \theta = x \sin \theta$ and $2x \sec \theta - y \operatorname{cosec} \theta = 3$, then what is $x^2 + 4y^2$ equal to:

[CDS-2016-2]

- (a) 1 (b) 2
(c) 4 (d) 8

87. If $\sin \theta + \cos \theta = \frac{1+\sqrt{3}}{2}$ where $0 < \theta < \frac{\pi}{2}$, then what is $\tan \theta + \cot \theta$ equal to?

[CDS-2016-2]

- (a) $\frac{\sqrt{3}}{4}$ (b) $\frac{1}{\sqrt{3}}$
(c) $\sqrt{3}$ (d) $\frac{4}{\sqrt{3}}$

88. If $A = \sin^2 \theta + \cos^4 \theta$ where $0 < \theta < \pi/2$, then which one of the following is correct?

[CDS-2016-2]

- (a) $1 \leq A \leq 2$ (b) $\frac{3}{4} \leq A \leq 1$
(c) $\frac{13}{16} \leq A \leq 2$ (d) $\frac{3}{4} \leq A \leq \frac{13}{16}$

89. What is $\frac{\cot A + \operatorname{cosec} A - 1}{\cot A - \operatorname{cosec} A + 1}$ equal to?

[CDS-2016-2]

- (a) $\frac{1+\cos A}{\sin A}$ (b) $\frac{1-\cos A}{\sin A}$
(c) $\frac{1+\sin A}{\cos A}$ (d) $\frac{1-\sin A}{\cos A}$

90. Consider the following:

1. $\sin 1^\circ > \sin 1^c$
2. $\cos 1^\circ < \cos 1^c$
3. $\tan 1^\circ > \tan 1^c$

Which of the above are not correct?

[CDS-2016-2]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

91. If $\tan^2 x + \frac{1}{\tan^2 x} = 2$ and $0^\circ < x < 90^\circ$, then what is the value of x ?

[CDS-2016-2]

- (a) 15° (b) 30°
(c) 45° (d) 60°

92. Consider the following:

1. $\frac{\cos 75^\circ}{\sin 15^\circ} + \frac{\sin 12^\circ}{\cos 78^\circ} - \frac{\cos 18^\circ}{\sin 72^\circ} = 1$
2. $\frac{\cos 35^\circ}{\sin 55^\circ} - \frac{\sin 11^\circ}{\cos 79^\circ} + \cos 28^\circ \operatorname{cosec} 62^\circ = 1$
3. $\frac{\sin 80^\circ}{\cos 10^\circ} - \sin 59^\circ \sec 31^\circ = 1$

Which of the above are correct?

[CDS-2016-2]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

93. What is the value of $\tan 1^\circ \tan 2^\circ \tan 3^\circ \tan 4^\circ \dots \tan 89^\circ$?

[CDS-2016-2]

- (a) 0 (b) 1
(c) 2 (d) $\sqrt{3}$

94. What is the minimum value of: $9\tan^2 \theta + 4\cot^2 \theta$?

[CDS-2017-1]

- (a) 6 (b) 9
(c) 12 (d) 13

95. If $x \sin \theta = y \cos \theta = \frac{2z \tan \theta}{1 - \tan^2 \theta}$, then what is $4z^2 (x^2 + y^2)$ equal to:

[CDS-2017-1]

- (a) $(x^2 + y^2)^3$ (b) $(x^2 - y^2)^2$
(c) $(x^2 - y^2)^2$ (d) $(x^2 + y^2)^2$

96. If $\cos \theta_1 + \cos \theta_2 + \cos \theta_3 = 3$, then what is $\sin \theta_1 + \sin \theta_2 + \sin \theta_3$ equal to?

[CDS-2017-1]

- (a) 0 (b) 1
(c) 2 (d) 3

97. What is the value of θ which satisfies the equation $\cos \theta + \tan \theta = 1$?

[CDS-2017-1]

- (a) 0° (b) 1
(c) 45° (d) 60°

98. What is the value of:

$$\sin x \sqrt{\frac{1}{1 + \cos x} + \frac{1}{1 - \cos x}} ?$$

[CDS-2017-1]

- (a) $\sqrt{2}$ (b) $2\sqrt{2}$
(c) $\sqrt{2} \tan x$ (d) 0

99. What is $\frac{\cos^4 A - \sin^4 A}{\cos^2 A - \sin^2 A}$ equal to?

[CDS-2017-1]

- (a) $\cos^2 A - \sin^2 A$ (b) $\cos A - \sin A$
(c) 1 (d) 2

100. If $7\sin^2 x + 3\cos^2 x = 4$, $0 < x < 90^\circ$, then what is the value of $\tan x$?

[CDS-2017-1]

- (a) $\sqrt{2}$ (b) 1
(c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{\sqrt{3}}$

101. If D is the number of degrees and R is the number of radians in an angle θ , then which one of the following is correct?

[CDS-2017-1]

- (a) $\pi D = 180R$ (b) $\pi D = 90R$
(c) $\pi R = 180D$ (d) $\pi R = 90D$

102. If $x = a \cos \theta + b \sin \theta$ and $y = a \sin \theta - b \cos \theta$, then what is $x^2 + y^2$ equal to?

[CDS-2017-1]

- (a) $2ab$ (b) $a + b$
(c) $a^2 + b^2$ (d) $a^2 - b^2$

103. If a triangle has sides 5, 13 and 12 units and θ is the acute angle of the triangle, then what is the value of $(\sin \theta + \cos \theta)$?

[CDS-2017-2]

- (a) $\frac{5}{13}$ (b) $\frac{7}{13}$
(c) $\frac{12}{13}$ (d) $\frac{17}{13}$

104. If $0 < x < \pi/2$, then $(\sin x + \operatorname{cosec} x)$ is:

[CDS-2017-2]

- (a) > 2 (b) < 2
(c) ≥ 2 (d) ≤ 2

105. If $\sin \theta = \frac{m^2 - n^2}{m^2 + n^2}$ and $0 < \theta < \frac{\pi}{2}$, then what is value of $\cos \theta$?

[CDS-2017-2]

- (a) $\frac{2mn}{m^2 + n^2}$ (b) $\frac{2mn}{m^2 - n^2}$
(c) $\frac{m^2 + n^2}{2mn}$ (d) $\frac{m^2 - n^2}{2mn}$

106. If $A = \frac{\sin 45^\circ - \sin 30^\circ}{\cos 45^\circ + \cos 60^\circ}$ and $B = \frac{\sec 45^\circ - \tan 45^\circ}{\operatorname{cosec} 45^\circ + \cot 45^\circ}$ then what one of the following is correct?

[CDS-2017-2]

- (a) $A = B$ (b) $A > B > 0$
(c) $A < B$ (d) $B < A < 0$

107. Consider the following statements:

1. If $45^\circ < \theta < 60^\circ$, then $\sec^2 \theta + \operatorname{cosec}^2 \theta = \alpha^2$ for some real number $\alpha > 1$.

2. If $0^\circ < \theta < 45^\circ$, then $\frac{1 + \cos \theta}{1 - \cos \theta} = x^2$ for some real number $x > 2$.

3. If $0^\circ < \theta < 45^\circ$, then $\frac{\cos \theta}{1 - \tan \theta} + \frac{\sin \theta}{1 - \cos \theta} \geq 2$

What is the number of true statement?

[CDS-2017-2]

- (a) Zero (b) One
(c) Two (d) Three

108. The value of $\frac{\sin 1^\circ}{\sin 1^\circ}$ where 1° represent 1 radian is:

[CDS-2017-2]

- (a) Equal to 1
(b) Less than 1
(c) Greater than 1 but less than 2
(d) Greater than 2

109. If θ measured in radians is the angle between the hour hand and the minute hand of a clock when the time is 4:36pm then which one of the following is correct?

[CDS-2017-2]

- (a) $\frac{3\pi}{5} < \theta < \frac{4\pi}{5}$ (b) $\frac{2\pi}{5} < \theta < \frac{3\pi}{5}$
(c) $\frac{\pi}{5} \leq \theta \leq \frac{2\pi}{5}$ (d) $\frac{7\pi}{15} \leq \theta \leq \frac{8\pi}{15}$

110. What is $\sin^4 \theta - \cos^4 \theta$ equal to for any real number θ ?

[CDS-2018-1]

- (a) 1 (b) $1 - 2 \sin^2 \theta$
(c) $2 \cos^2 \theta + 1$ (d) $1 - 2 \cos^2 \theta$

111. What is $\cot 1^\circ \cot 23^\circ \cot 45^\circ \cot 67^\circ \cot 89^\circ$ equal to?

[CDS-2018-1]

- (a) 0 (b) 1
(c) $\frac{1}{2}$ (d) $\frac{1}{3}$

112. Consider the following statements:

1. $(\sec^2 \theta - 1)(1 - \operatorname{cosec}^2 \theta) = 1$

2. $\sin \theta (1 + \cos \theta)^{-1} + (1 + \cos \theta) (\sin \theta)^{-1} = 2 \operatorname{cosec} \theta$

Which of the above is/are correct?

[CDS-2018-1]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

113. If $\sec x \operatorname{cosec} x = 2$, then what is $\tan^n x + \cot^n x$ equal to?

[CDS-2018-1]

- (a) 2 (b) 2^{n+1}
(c) 2^n (d) 2^{n-1}

114. If $\cos x + \cos^2 x = 1$, then what is $\sin^2 x + \sin^4 x$ equal to?

[CDS-2018-1]

- (a) 1 (b) 1.5
(c) 2 (d) 3

115. If $\sin A + \cos A = p$ and $\sin^3 A + \cos^3 A = q$, then which one of the following is correct?

[CDS-2018-1]

- (a) $p^3 - 3p + q = 0$ (b) $q^3 - 3q + 2p = 0$
(c) $p^3 - 3p + 2q = 0$ (d) $p^3 + 3p + 2q = 0$

116. If $x = \frac{\sec^2 \theta - \tan \theta}{\sec^2 \theta + \tan \theta}$, then which one of the following is correct?

[CDS-2018-1]

- (a) $\frac{1}{3} < x < 3$ (b) $x \in \left[\frac{1}{3}, 3\right]$
(c) $-3 < x < -\frac{1}{3}$ (d) $\frac{1}{3} \leq x \leq 3$

117. ABC is a right angled triangle with base BC and height AB. The hypotenuse AC is four times the length of the perpendicular drawn to it from the opposite vertex. What is $\tan C$ equal to?

[CDS-2018-1]

- (a) $2 - \sqrt{3}$ (b) $\sqrt{3} - 1$
(c) $2 + \sqrt{3}$ (d) $\sqrt{3} + 1$

118. What angle does the hour hand of a clock describe in 10 minutes of time?

[CDS-2018-1]

- (a) 1° (b) 5°
(c) 6° (d) 10°

119. If $\cos \theta = \frac{1}{\sqrt{5}}$, where $0 < \theta < \frac{\pi}{2}$, then

$\frac{2 \tan \theta}{1 - \tan^2 \theta}$ is equal to:

[CDS-2018-2]

- (a) $\frac{4}{3}$ (b) $-\frac{4}{3}$
(c) $\frac{1}{3}$ (d) $-\frac{2}{3}$

120. If $0 < \theta < 90^\circ$, $0 < \phi < 90^\circ$ and $\cos \theta < \cos \phi$, then which one of the following is correct?

[CDS-2018-2]

- (a) $\theta < \phi$
- (b) $\theta > \phi$
- (c) $\theta + \phi = 90^\circ$
- (d) No conclusion can be drawn

121. Let $\sin(A + B) = \frac{\sqrt{3}}{2}$ and $\cos B = \frac{\sqrt{3}}{2}$, where A, B are acute angles. What is $\tan(2A - B)$ equal to?

[CDS-2018-2]

- (a) $\frac{1}{2}$
- (b) $\sqrt{3}$
- (c) $\frac{1}{\sqrt{3}}$
- (d) 1

122. Consider the following statements:

1. If $\frac{\cos \theta}{1 - \sin \theta} + \frac{\cos \theta}{1 + \sin \theta} = 4$, where $0 < \theta < 90^\circ$, then $\theta = 60^\circ$

2. If $3 \tan \theta + \cot \theta = 5 \operatorname{cosec} \theta$, where $0 < \theta < 90^\circ$, then $\theta = 60^\circ$

Which of the statements given above is/are correct?

[CDS-2018-2]

- (a) 1 Only
- (b) 2 Only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

123. Consider the following statements:

1. $\cos^2 \theta = 1 - \frac{p^2 + q^2}{2pq}$, where p, q are non-zero real numbers, is possible only when $p = q$.

2. $\tan^2 \theta = \frac{4pq}{(p+q)^2} - 1$, where p, q are non-zero real numbers, is above is/are correct?

[CDS-2018-2]

- (a) 1 Only
- (b) 2 Only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

124. Consider the following statements:

1. $\cos \theta + \sec \theta$ can never be equal to 1.5

2. $\sec^2 \theta + \operatorname{cosec}^2 \theta$ can never be less than 4

Which of the statements given above is/are correct?

[CDS-2018-2]

- (a) 1 Only
- (b) 2 Only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

125. If $\sin^2 x + \sin x = 1$, when what is the value of $\cos^{12} x + 3 \cos^{10} x + 3 \cos^8 x + \cos^6 x$?

[CDS-2018-2]

- (a) -1
- (b) 0
- (c) 1
- (d) 8

126. If $3 \sin \theta + 5 \cos \theta = 4$, then what is the value of $(3 \cos \theta - 5 \sin \theta)^2$?

[CDS-2018-2]

- (a) 9
- (b) 12
- (c) 16
- (d) 18

127. If $\cot \theta (1 + \sin \theta) = 4m$ and $\cot \theta (1 - \sin \theta) = 4n$, then which one of the following is correct?

[CDS-2018-2]

- (a) $(m^2 + n^2) = mn$
- (b) $(m^2 - n^2)^2 = mn$
- (c) $(m^2 - n^2)^2 = m^2 n^2$
- (d) $(m^2 + n^2)^2 = m^2 n^2$

128. The minute hand of a clock overtakes the hour hand after every 72 minutes of correct time. How much time does the clock lose or gain in a day of normal time?

[CDS-2018-2]

- (a) lose $121 \frac{9}{11}$ minutes
- (b) lose $157 \frac{1}{11}$ minutes
- (c) gain $121 \frac{9}{11}$ minutes
- (d) gain $157 \frac{1}{11}$ minutes

129. If $\tan x = 1$, $0 < x < 90^\circ$, then what is the value of $2 \sin x \cos x$?

[CDS-2019-1]

- (a) $\frac{1}{2}$
- (b) 1
- (c) $\frac{\sqrt{3}}{2}$
- (d) $\sqrt{3}$

130. What is the value of $\sin 46^\circ \cos 44^\circ + \cos 46^\circ \sin 44^\circ$?

[CDS-2019-1]

- (a) $\sin 2^\circ$
- (b) 0
- (c) 1
- (d) 2

131. Suppose $0 < \theta < 90^\circ$, then for every θ , $4 \sin^2 \theta + 1$ is greater than or equal to:

[CDS-2019-1]

- (a) 2
- (b) $4 \sin \theta$
- (c) $4 \cos \theta$
- (d) $4 \tan \theta$

132. What is the value of $\tan 1^\circ \tan 2^\circ \tan 3^\circ \dots \tan 89^\circ$?

[CDS-2019-1]

- (a) 0
- (b) 1
- (c) 2
- (d) ∞

- 133.** There are two parallel streets each directed North to South. A person in the first street travelling from South to North wishes to take the second street which is on his right side. At some place, he makes at 150° turn to the right and he travels for 15 minutes at the speed of 20 km/hr. After that he takes a left turn of 60° and travels for 20 minutes at the speed of 30 km/hr in order to meet the second street. What is the distance between the two streets?
[CDS-2019-1]
(a) 7.5km (b) 10.5km
(c) 12.5km (d) 15km
- 134.** If $3 \tan \theta = \cot \theta$ where $0 \leq \theta \leq \pi/2$, then what is the value of θ ?
[CDS-2019-1]
(a) $\pi/6$ (b) $\pi/4$
(c) $\pi/3$ (d) $\pi/2$
- 135.** What is the value of $\sin^2 25^\circ + \sin^2 65^\circ$?
[CDS-2019-1]
(a) 0 (b) 1
(c) 2 (d) 4
- 136.** What is the value of $\sin^6 \theta + \cos^6 \theta + 3 \sin^2 \theta \cos^2 \theta - 1$?
[CDS-2019-1]
(a) 0 (b) 1
(c) 2 (d) 4
- 137.** Consider the following for real numbers α , β , γ and δ :
[CDS-2019-1]
1. $\sec \alpha = \frac{1}{4}$
2. $\tan \beta = 20$
3. $\operatorname{cosec} \gamma = \frac{1}{2}$
4. $\cos \delta = 2$
How many of the above statements are not possible
(a) One (b) Two
(c) Three (d) Four
- 138.** If $\cos^2 x + \cos x = 1$, then what is the value of $\sin^{12} x + 3 \sin^{10} x + 3 \sin^8 x + \sin^6 x$?
[CDS-2019-2]
(a) 1 (b) 2
(c) 4 (d) 8
- 139.** If $0 < \theta < 90^\circ$, $\sin \theta = 3/5$ and $x = \cot \theta$, then what is the value of:
 $1 + 3x + 9x^2 + 27x^3 + 81x^4 + 243x^5$?
[CDS-2019-2]
(a) 941 (b) 1000
- (c) 1220 (d) 1365
- 140.** What is The value of $\frac{\sin 19^\circ}{\cos 71^\circ} + \frac{\cos 73^\circ}{\sin 17^\circ}$?
[CDS-2019-2]
(a) 0 (b) 1
(c) 2 (d) 4
- 141.** $\log_{10} (\cos \theta) + \log_{10} (\sin \theta) + \log_{10} (\tan \theta) + \log_{10} (\cot \theta) + \log_{10} (\sec \theta) + \log_{10} (\operatorname{cosec} \theta)$?
[CDS-2019-2]
(a) -1 (b) 0
(c) 0.5 (d) 1
- 142.** A wheel makes 360 revolutions in one minute. What is the number of radians it turns in one second?
[CDS-2020-1]
(a) 4π (b) 6π
(c) 12π (d) 16π
- 143.** What is the least value of $(25 \operatorname{cosec}^2 x + \sec^2 x)$?
[CDS-2020-1]
(a) 40 (b) 36
(c) 26 (d) 24
- 144.** Let $0 < \theta < 90^\circ$ and $100\theta = 90^\circ$. If $\alpha = \prod_{n=1}^{99} \cot n\theta$, then which one of the following is correct?
[CDS-2020-1]
(a) $\alpha = 1$ (b) $\alpha = 0$
(c) $\alpha > 1$ (d) $0 < \alpha < 1$
- 145.** If $\tan 6\theta = \cot 2\theta$, where $0 < 6\theta < \pi/2$, then what is the value of $\sec 4\theta$?
[CDS-2020-1]
(a) $\sqrt{2}$ (b) 2
(c) $\frac{2}{\sqrt{3}}$ (d) $\frac{4}{3}$
- 146.** What is the value of $\sin^2 6^\circ + \sin^2 12^\circ + \sin^2 18^\circ + \dots + \sin^2 84^\circ + \sin^2 90^\circ$?
[CDS-2020-1]
(a) 1 (b) 2
(c) 4 (d) 8
- 147.** What is $\frac{\cos \theta}{1 + \sin \theta} + \frac{1}{\cot \theta}$ equal to?
[CDS-2020-1]
(a) $\operatorname{cosec} \theta$ (b) $\sec \theta$
(c) $\sec \theta + \operatorname{cosec} \theta$ (d) $\operatorname{cosec} \theta - \cot \theta$

148. What is $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} - \frac{\sin \theta + 1}{\cos \theta}$ equal to?

[CDS-2020-1]

- (a) 0 (b) 1
(c) $2 \sin \theta$ (d) $2 \cos \theta$

149. What is $(\tan x + \tan y)(1 - \cot x \cot y) + (\cot x + \cot y)(1 - \tan x \tan y)$ equal to?

[CDS-2020-1]

- (a) 0 (b) 1
(c) 2 (d) 4

150. What is $\sqrt{\frac{\sec x - \tan x}{\sec x + \tan x}}$ equal to?

[CDS-2020-1]

- (a) $\frac{1}{\sin x + \cos x}$ (b) $\frac{1}{\tan x + \cot x}$
(c) $\frac{1}{\sec x + \tan x}$ (d) $\frac{1}{\operatorname{cosec} x + \cot x}$

151. If θ lies in the first quadrant and $\cot \theta = \frac{63}{16}$, then what is the value of $(\sin \theta + \cos \theta)$?

[CDS-2020-1]

- (a) 1 (b) $\frac{69}{65}$
(c) $\frac{79}{65}$ (d) 2

152. What is the value of $\frac{1 - 2 \sin^2 \theta \cos^2 \theta}{\sin^4 \theta + \cos^4 \theta} + 4$ equal to?

[CDS-2020-1]

- (a) 0 (b) 1
(c) 2 (d) 5

153. A rectangle is 48cm long and 14 cm wide. If the diagonal makes an angle θ with the longer side, then what is $(\sec \theta + \operatorname{cosec} \theta)$ equal to?

[CDS-2020-1]

- (a) $\frac{775}{168}$ (b) $\frac{725}{168}$
(c) $\frac{375}{84}$ (d) $\frac{325}{84}$

154. If $\operatorname{cosec} \theta - \sin \theta = m$ and $\sec \theta - \cos \theta = n$, then what is $m^{\frac{4}{3}} n^{\frac{2}{3}} + m^{\frac{2}{3}} n^{\frac{4}{3}}$ equal to?

[CDS-2020-2]

- (a) 0 (b) 1
(c) mn (d) $m^2 n^2$

155. If $\cos \theta + \sec \theta = k$, then what is the value of $\sin^2 \theta - \tan^2 \theta$?

[CDS-2020-2]

- (a) $4-k$ (b) $4-k^2$
(c) k^2-4 (d) k^2+2

156. ABC is a triangle inscribed in a semicircle of diameter AB. What is $\cos(A+B) + \sin(A+B)$ equal to?

[CDS-2020-2]

- (a) 0 (b) $1/4$
(c) $1/2$ (d) 1

157. Consider the following statements:

1. $\sin \theta = x + 1/x$ is possible for some real value of x .

2. $\cos \theta = x + 1/x$ is possible for some real value of x

Which of the above statements is/are correct?

[CDS-2020-2]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

158. What is the magnitude (in radian) of the interior angle of a regular pentagon?

[CDS-2020-2]

- (a) $\frac{\pi}{5}$ (b) $\frac{2\pi}{5}$
(c) $\frac{3\pi}{5}$ (d) $\frac{4\pi}{5}$

159. The difference between two angles is 15° and the sum of the angles in radian is $\frac{5\pi}{12}$.

The bigger angle is k times the smaller angle. What is k equal to?

[CDS-2020-2]

- (a) $\frac{4}{3}$ (b) $\frac{3}{2}$
(c) $\frac{6}{5}$ (d) $\frac{7}{6}$

160. Consider the following statements:

1. The equation $2 \sin^2 \theta - \cos \theta + 4 = 0$ is possible for all θ .

2. $\tan \theta + \cot \theta$ cannot be less than 2, where $0 < \theta < \pi/2$.

Which of the above statements is/are correct?

[CDS-2020-2]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) 2

161. What is the maximum value of $3 \sin \theta - 4$?

[CDS-2020-2]

- (a) -4 (b) -1
(c) 0 (d) 1

162. If $\sin \theta + \cos \theta = \sqrt{2}$, then what is $\sin^6 \theta + \cos^6 + 6\sin^2 \theta \cos^2 \theta$ equal to?

[CDS-2020-2]

- (a) $\frac{1}{4}$ (b) $\frac{3}{4}$
(c) 1 (d) $\frac{7}{4}$

163. What is the least value of $9 \sin^2 \theta + 16 \cos^2 \theta$?

[CDS-2020-2]

- (a) 0 (b) 9
(c) 16 (d) 26

164. If $\cos 47^\circ + \sin 47^\circ = k$, then what is the value of $\cos^2 47^\circ - \sin^2 47^\circ$?

[CDS-2020-2]

- (a) $k\sqrt{2-k^2}$ (b) $-k\sqrt{2-k^2}$
(c) $k\sqrt{1-k^2}$ (d) $-k\sqrt{1-k^2}$

165. If $\operatorname{cosec} \theta - \sin \theta = p^3$ and $\sec \theta - \cos \theta = q^3$, then what is the value of $\tan \theta$?

[CDS-2020-2]

- (a) $\frac{p}{q}$ (b) $\frac{q}{p}$
(c) pq (d) p^2q^2

166. If $0 \leq \alpha \leq \beta \leq 90^\circ$ such that $\cos(\alpha - \beta) = 1$, then what is $\sin \alpha - \sin \beta + \cos \alpha - \cos \beta$ equal to?

[CDS-2020-2]

- (a) -1 (b) 0
(c) 1 (d) 2

167. Consider the following statements:

1. The value of $\cos 61^\circ + \sin 29^\circ$ cannot exceed 1.
 2. The value of $\tan 23^\circ - \cot 67^\circ$ is less than 0.
- Which of the above statements is/are correct?

[CDS-2020-2]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

168. Consider the following inequalities:

1. $\sin 1^\circ < \cos 57^\circ$
 2. $\cos 60^\circ > \sin 57^\circ$
- Which of the above is/are correct?

[CDS-2021-1]

- (a) 1 Only (b) 2 Only

- (c) Both 1 and 2 (d) Neither 1 nor 2

169. If $p = \sec \theta - \tan \theta$ and $q = \operatorname{cosec} \theta + \cot \theta$, then what is $p + q$ ($p-1$) equal to?

[CDS-2021-1]

- (a) -1 (b) 0
(c) 1 (d) 2

170. If $\operatorname{cosec} \theta - \cot \theta = m$, then what is $\operatorname{cosec} \theta$ equal to?

[CDS-2021-1]

- (a) $m + \frac{1}{m}$ (b) $m - \frac{1}{m}$
(c) $\frac{m}{2} + \frac{2}{m}$ (d) $\frac{m}{2} + \frac{1}{2m}$

171. Let ABC be a triangle right angled at C, then what is $\tan A + \tan B$ equal to?

[CDS-2021-1]

- (a) $\frac{a}{bc}$ (b) $\frac{a^2}{bc}$
(c) $\frac{b^2}{ca}$ (d) $\frac{c^2}{ab}$

172. Let $\cos \alpha + \cos \beta = 2$ and $\sin \alpha + \sin \beta = 0$, where $0 \leq \alpha \leq 90^\circ$, $0 \leq \beta \leq 90^\circ$. What is the value of $\cos 2\alpha - \cos 2\beta$?

[CDS-2021-1]

- (a) 0
(b) 1
(c) 2
(d) Cannot be determined due to insufficient data

173. If $\sec \theta + \cos \theta = \frac{5}{2}$, where $0 \leq \theta \leq 90^\circ$, then what is the value of $\sin^2 \theta$?

[CDS-2021-1]

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{3}{4}$ (d) 1

174. What is $(1 + \cot \theta - \operatorname{cosec} \theta)(1 + \tan \theta + \sec \theta)$ equal to?

[CDS-2021-1]

- (a) 4 (b) 3
(c) 2 (d) 1

175. If $6 + 8 \tan \theta = \sec \theta$ and $8 - 6 \tan \theta = k \sec \theta$, then what is the value of k^2 ?

[CDS-2021-1]

- (a) 11 (b) 22
(c) 77 (d) 99

- 176.** What is the least value of $3\sin^2 \theta + 4 \cos^2 \theta$?
[CDS-2021-1]
(a) 5 (b) 4
(c) 3 (d) 2
- 177.** If $\sin \theta \cos \theta = k$, where $0 \leq \theta \leq \pi/2$, then which one of the following is correct?
[CDS-2021-1]
(a) $0 \leq k \leq 1$ (b) $0 \leq k \leq 0.5$ only
(c) $0.5 \leq k \leq 1$ only (d) $0 < k < 1$
- 178.** If $p = \sin^2 \theta + \cos^4 \theta$ for $0 \leq \theta \leq \pi/2$, then consider the following statements:
1. p can be less than $3/4$
2. p can be more than 1
Which of the above statements is/are correct?
[CDS-2021-1]
(a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2
- 179.** What is the ratio of the greatest to the smallest value of $2 - 2 \sin x - \sin^2 x$, $0 \leq x \leq \pi/2$?
[CDS-2021-1]
(a) -3 (b) -1
(c) 1 (d) 3
- 180.** If the equation $x^2 + y^2 - 2xy \sin^2 \theta = 0$ contains real solution for x and y , then:
[CDS-2021-1]
(a) $x = y$ (b) $x = -y$
(c) $x = 2y$ (d) $2x = y$
- 181.** Between 3 and 4 O' clock, both hour hand and minute hand will coincide past 3 O'clock between
[CDS-2021-2]
(a) 15-16 minutes (b) 16-17 minutes
(c) 17-18 minutes (d) 18-19 minutes
- 182.** What is the minimum value of $\cos^3 \theta + \sec^3 \theta$, where $0^\circ \leq \theta < 90^\circ$?
[CDS-2021-2]
(a) 0 (b) 1
(c) 2 (d) None of these
- 183.** If $14\sin^2 \theta + 10\cos^2 \theta = 11$, where $0^\circ < \theta < 90^\circ$, then what is the value of $\tan \theta + \cot \theta$?
[CDS-2021-2]
(a) $\frac{4}{\sqrt{3}}$ (b) $\frac{2}{\sqrt{3}}$
(c) $\sqrt{3}$ (d) $2\sqrt{3}$
- 184.** What is $\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} + \frac{\sin^3 \theta - \cos^3 \theta}{\sin \theta - \cos \theta}$ equal to?
[CDS-2021-2]
(a) 0 (b) 1
(c) 2 (d) 4
- 185.** What is the maximum value of $1 + 2\sin^2 \theta \cos^2 \theta - \sin^4 \theta - \cos^4 \theta$, where $0^\circ < \theta < 90^\circ$?
[CDS-2021-2]
(a) 1 (b) 2
(c) 3 (d) 4
- 186.** If $\frac{\cos^2 \theta - 3\cos \theta + 2}{\sin^2 \theta} = 1$, where $0^\circ < \theta < 90^\circ$ then what is $\sin^2 \theta + \cos \theta$ equal to?
[CDS-2021-2]
(a) $5/4$ (b) $3/2$
(c) $7/4$ (d) 2
- 187.** Consider the following:
1. $\sin^4 \theta - \sin^2 \theta = \cos^4 \theta - \cos^2 \theta$
2. $\sin^4 \theta + \cos^4 \theta = 1 + 2\sin^2 \theta \cos^2 \theta$
3. $\tan^4 \theta + \tan^2 \theta = \sec^4 \theta - \sec^2 \theta$
Which of the above are identities?
[CDS-2021-2]
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3
- 188.** What is the value of $\sin 24^\circ \sin 66^\circ - \cos 24^\circ \cos 66^\circ + \tan 24^\circ \tan 66^\circ - \cot 24^\circ \cot 66^\circ$?
[CDS-2021-2]
(a) 0 (b) 1
(c) 2 (d) 3
- 189.** If $x = p \sin A \cos B$, $y = p \sin A \sin B$ and $z = p \cos A$, then what is the value of $x^2 + y^2 + z^2$?
[CDS-2021-2]
(a) $-p^2$ (b) 0
(c) p^2 (d) $2p^2$
- 190.** If $x = m \sec A + n \tan A$ and $y = m \tan A + n \sec A$, then what is $x^2 - y^2$ equal to?
[CDS-2021-2]
(a) $m^2 - n^2$ (b) $m^2 + n^2$
(c) $m^2 + n^2 - mn$ (d) $m^2 - n^2 + mn$
- 191.** If for some θ lying between 0° and 90° , $\tan \theta = 1$, then what is the value of $\sin^2 \theta - 2 \sin \theta \cos \theta$?
[CDS-2021-2]
(a) -1 (b) 0
(c) $1/2$ (d) $-1/2$

192. What is $\frac{2\sin^3 \theta - \sin \theta}{\cos \theta - 2\cos^3 \theta}$, ($0^\circ < \theta < 90^\circ$) equal to ?

[CDS-2021-2]

- (a) $\sin \theta$ (b) $\cos \theta$
(c) $\tan \theta$ (d) $\cot \theta$

193. If A, B and C are interior angles of a triangle ABC, then what is $\tan\left(\frac{B+C}{2}\right) + \sin\left(\frac{B+C}{2}\right) - \cot\left(\frac{A}{2}\right) - \cos\left(\frac{A}{2}\right)$ equal to?

[CDS-2021-2]

- (a) 0 (b) $1/2$
(c) $\sin\left(\frac{A+B+C}{2}\right)$ (d) $\tan\left(\frac{A+B+C}{2}\right)$

194. In a triangle ABC, right-angled at B, AB + BC = $10(1+\sqrt{3})$ cm and length of hypotenuse is 20 cm. What is the value of $\tan A + \cot A$?

[CDS-2021-2]

- (a) $\frac{4}{\sqrt{3}}$ (b) $\frac{2}{\sqrt{3}}$
(c) $\sqrt{3}$ (d) $2\sqrt{3}$

195. If A, B, C are acute angles and $\sin(B+C-A) = \cos(C+A-B) = \tan(A+B-C) = 1$ then what is $(A+B+C)$ equal to?

[CDS-2022-1]

- (a) 90° (b) 120°
(c) 135° (d) 150°

196. If A, B, C, D are the angles of a cyclic quadrilateral, then what is the value of the following?

$$\sin\left(\frac{A+C}{2}\right) + \sin\left(\frac{B+D}{2}\right)$$

[CDS-2022-1]

- (a) 2 (b) 1
(c) 0 (d) -1

197. What is $(\sec^2 \alpha + \tan \alpha \cdot \tan \beta - \tan^2 \alpha)^2 + (\tan \alpha - \tan \beta)^2 - \sec^2 \alpha \cdot \sec^2 \beta$ equal to?

[CDS-2022-1]

- (a) -1 (b) 0
(c) 1 (d) 2

198. If $\tan \theta + \sec \theta = 3$, then what is the value of $3\tan \theta + 9\sec \theta$?

[CDS-2022-1]

- (a) 15 (b) 17
(c) 19 (d) 21

199. Consider the following:

$$1. \sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta} = \tan \theta + \cot \theta, \text{ where } 0 < \theta < 90^\circ$$

$$2. \sqrt{\tan^2 \theta + \cot^2 \theta + 4} = \sec \theta + \operatorname{cosec} \theta, \text{ where } 0 < \theta < 90^\circ.$$

Which of the above is/are an identity/identities?

[CDS-2022-1]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

200. What is the length of the chord of a unit circle which subtends an angle 2θ at the centre, where $\theta < 45^\circ$?

[CDS-2022-1]

- (a) $\sin 2\theta$ (b) $\cos 2\theta$
(c) $2\sin \theta$ (d) $2\cos \theta$

201. If $\tan^2 \theta + 3\sec \theta - 9 = 0$ where $0 < \theta < 90^\circ$, then what is the value of $12\cot^2 \theta + 3\operatorname{cosec} \theta$?

[CDS-2022-1]

- (a) $(\sqrt{3} + 1)^2$ (b) $(\sqrt{3} + 2)^2$
(c) $(2\sqrt{3} + 1)^2$ (d) $(3\sqrt{3} + 1)^2$

202. If $\frac{\cos \theta}{\operatorname{cosec} \theta + 1} + \frac{\cos \theta}{\operatorname{cosec} \theta - 1} = 2$ where $0 < \theta < 90^\circ$, then what is the value of $\sin^4 \theta + \cos^4 \theta$?

[CDS-2022-1]

- (a) 2 (b) 1
(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

203. If $x = \frac{1 - \cos \theta + \sin \theta}{1 + \sin \theta}$ then what is $\frac{\sin \theta + \cos \theta - 1}{\cos \theta}$ equal to?

[CDS-2022-1]

- (a) $\frac{1}{x}$ (b) x
(c) $1 + x$ (d) $x - 1$

204. If $\cos(x+y) = 0$ and $\sin(x-y) = \frac{1}{2}$, where $x, y \in \left[0, \frac{\pi}{2}\right]$, then what is the value of $\cot(2x - y)$?

[CDS-2022-1]

- (a) 0 (b) $\frac{1}{2}$
(c) 1 (d) 2

205. What is the minimum value of $\sin^4 \theta + \cos^4 - 2\sin^2 \theta \cos^2 \theta$?

[CDS-2022-1]

- (a) 0
(b) 1

- (c) 1
(d) Minimum value does not exist

206. If $\cos\theta + \sec\theta - 2 = 0$, where $0 \leq \theta < \frac{\pi}{2}$, then what is the value of $\cos^4\theta + \sec^4\theta - 2$?

[CDS-2022-1]

- (a) -2 (b) -1
(c) 0 (d) 2

207. If $y = \cos^2 x + \sec^2 x$, where $0 \leq x < \frac{\pi}{2}$, then which one of the following is correct?

[CDS-2022-1]

- (a) $0 < y < 0.5$ (b) $0.5 \leq y < 1$
(c) $1 \leq y < 2$ (d) $y \geq 2$

208. What is the value of the following?

$$\frac{2 \sin 68^\circ}{\cos 22^\circ} - \frac{2 \cot 15^\circ}{5 \tan 75^\circ} - \frac{3 \tan 20^\circ \tan 40^\circ \tan 45^\circ \tan 50^\circ \tan 70^\circ}{5}$$

[CDS-2022-1]

- (a) -1 (b) 0
(c) 1 (d) 5

209. What is the angle between the hour hand and the minute hand of a clock when the clock shows 4 hours 40 minutes?

[CDS 2022 (II)]

- (a) 80° (b) 100°
(c) 120° (d) 220°

210. In a triangle ABC, $\angle A = 2\theta$, $\angle B = C = 4\theta$ and θ satisfies $4 \sin^2 \theta + 2 \sin \theta - 1 = 0$. What is the ratio of BC to AB?

[CDS 2022 (II)]

- (a) $\sqrt{5} - 1$ (b) $\frac{(\sqrt{5} - 1)}{2}$
(c) $\frac{(\sqrt{5} - 1)}{4}$ (d) $2(\sqrt{5} - 1)$

211. If $\tan \theta = \frac{\sqrt{q^2 - p^2}}{p}$; $0 < \theta < 90^\circ$, then what is $\sec \theta + \cos \theta + 2$ equal to?

[CDS 2022 (II)]

- (a) $\frac{p^2 + q^2}{pq}$ (b) $\frac{(p+q)^2}{pq}$
(c) $\frac{(p+q)^2}{2pq}$ (d) $\frac{(p-q)^2}{pq}$

212. If $11 \sin \theta + 60 \cos \theta = 61$; $0 < \theta < 90^\circ$, then what is the value of $\sqrt{660(\tan \theta + \cot \theta)}$?

[CDS 2022 (II)]

- (a) 61 (b) $61\sqrt{2}$
(c) 122 (d) $122\sqrt{2}$

213. What is the diameter of a circle inscribed in a regular polygon of 15 sides with side length unity?

[CDS 2022 (II)]

- (a) $0.5 \cot 12^\circ$ (b) $\cot 12^\circ$
(c) $0.5 \tan 12^\circ$ (d) $\tan 12^\circ$

214. If $x = \frac{1 + \sin \theta}{\cos \theta}$, then what is $\frac{\tan \theta + \sec \theta - 1}{\tan \theta - \sec \theta + 1}$ equal to?

[CDS 2022 (II)]

- (a) -x (b) x
(c) 2x (d) $\frac{x}{2}$

215. Consider the following:

- $\frac{\tan \theta + \sin \theta}{\tan \theta - \sin \theta} = \frac{\sec \theta + 1}{\sec \theta - 1}$, where $0 < \theta < \frac{\pi}{2}$
- $\frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta} = \frac{2 \tan \theta}{\tan^2 \theta + 1}$, where $0 < \theta < \frac{\pi}{2}$

Which of the above is/are identities?

[CDS 2022 (II)]

- (a) only 1 (b) only 2
(c) both 1 and 2 (d) neither 1 nor 2

216. If $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = 5$, where $0 < \theta < \frac{\pi}{2}$, $\theta \neq \frac{\pi}{4}$, then what is the value of $\frac{2 \sin \theta + 3 \cos \theta}{3 \sin \theta - 2 \cos \theta}$?

[CDS 2022 (II)]

- (a) $\frac{8}{5}$ (b) 2
(c) $\frac{12}{5}$ (d) 3

217. If $\frac{1}{\operatorname{cosec} \theta - \cot \theta} - \frac{1}{\sin \theta} = x$, then what is $\frac{1}{\operatorname{cosec} \theta + \cot \theta} - \frac{1}{\sin \theta}$ equal to, where $0 < \theta < \frac{\pi}{2}$?

[CDS 2022 (II)]

- (a) -x (b) x
(c) $\frac{1}{x}$ (d) $-\frac{1}{x}$

218. What is $2 \sin^6 \theta + 2 \cos^6 \theta - 3 \sin^4 \theta - 3 \cos^4 \theta$ equal to?

[CDS 2022 (II)]

- (a) -1 (b) 0
(c) 1 (d) 2

219. What is the minimum value of $3600 \sec^2 \theta + 121 \operatorname{cosec}^2 \theta$, where $0 < \theta < \frac{\pi}{2}$?

[CDS 2022 (II)]

- (a) 1320 (b) 2401
(c) 3721 (d) 5041

220. What is $\sqrt{\left(\frac{1+\sin \theta}{\cos \theta}\right)^2 + \left(\frac{\cos \theta}{1+\sin \theta}\right)^2} - 2$ equal to, where $0 < \theta < \frac{\pi}{2}$?

[CDS 2022 (II)]

- (a) $\tan \theta$ (b) $\cot \theta$
(c) $2 \tan \theta$ (d) $2 \cot \theta$

221. In a triangle ABC, $\angle A = 60^\circ$. What is $AB^2 + AC^2 - BC^2$ equal to?

[CDS 2022 (II)]

- (a) AB.AC (b) AB.BC
(c) AC.BC (d) 2AB.AC

222. What is the value of $\frac{\cos^2 32^\circ + \cos^2 58^\circ}{\sec^2 50^\circ - \cot^2 40^\circ} + 4 \tan 13^\circ \tan 37^\circ \tan 53^\circ \tan 77^\circ$?

[CDS - 2023 (1)]

- (a) 2 (b) 3
(c) 4 (d) 5

223. What is the value of $(1 + \cot^2 \theta)(1 + \cos \theta)(1 - \cos \theta) - (1 + \tan^2 \theta)(1 + \sin \theta)(1 - \sin \theta)$?

[CDS - 2023 (1)]

- (a) -1 (b) 0
(c) 1 (d) 2

224. If $2 \cos^2 \theta + \sin \theta - 2 = 0$, $0 < \theta \leq \frac{\pi}{2}$, then what is the value of θ ?

[CDS - 2023 (1)]

- (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

225. Consider the following:

$$1. \cos^4 \theta - \sin^4 \theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}, 0 < \theta < \frac{\pi}{2}$$

$$2. \operatorname{cosec} \theta + \cot \theta = \frac{1}{\operatorname{cosec} \theta - \cot \theta}, 0 < \theta < \frac{\pi}{2}$$

$$3. \cos^2 \theta - \sin^2 \theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}, 0 < \theta < \frac{\pi}{2}$$

Which of the above equations are identities.

[CDS - 2023 (1)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

226. If $\sin \theta = \frac{12}{13}$, then what is the value of $(\tan \theta + \sec \theta)^2 (\operatorname{cosec} \theta - \cot \theta)^{-2}$, $0 < \theta < \frac{\pi}{2}$?

[CDS - 2023 (1)]

- (a) $\frac{121}{4}$ (b) $\frac{169}{9}$
(c) $\frac{196}{9}$ (d) $\frac{225}{4}$

227. If $\tan^8 \theta + \cot^8 \theta = m$, then what is the value of $\tan \theta + \cot \theta$?

[CDS - 2023 (1)]

- (a) $\sqrt{\sqrt{m+2}+2}$ (b) $\sqrt{\sqrt{m+4}+2}$
(c) $\sqrt{\sqrt{\sqrt{m+2}+2}+2}$ (d) $\sqrt{\sqrt{\sqrt{m+4}+2}+2}$

228. What is the minimum value of $6 - 4 \sin \theta$, $0 \leq \theta \leq \frac{\pi}{2}$?

[CDS - 2023 (1)]

- (a) 1 (b) 2
(c) 4 (d) 6

229. What is the value of x that satisfies $4 \cos^2 30^\circ + 2x \sin 30^\circ - \cot^2 30^\circ - 6 \tan 15^\circ \tan 75^\circ = 0$?

[CDS - 2023 (1)]

- (a) 1 (b) 2
(c) 3 (d) 6

230. Which of the following is/are identity/identities?

$$1. \frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} + \sin \theta \cos \theta = 1; 0 < \theta < \pi/2$$

$$2. 1 - \sin^6 \theta = \cos^2 \theta (\cos^4 \theta + 3 \sin^2 \theta)$$

Select the correct answer using the code given below:

[CDS-2023 (2)]

- (a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

231. If $7 \sin^4 \theta + 9 \cos^4 \theta + 42 \sin^2 \theta = 16$, $0 < \theta < \pi/2$, then what is $\tan \theta$ equal to?

[CDS-2023 (2)]

- (a) 1 (b) $\sqrt{2}$

(c) $\sqrt{3}$

(d) $\frac{1}{\sqrt{3}}$

232. If $\tan A + \cot A = 2$, where $0 < A < 90^\circ$, then what is the value of $\tan^2 A + \tan^3 A + \tan^4 A + \dots + \tan^n A$?

[CDS-2023 (2)]

(a) 1

(b) $n-2$

(c) $n-1$

(d) n

233. If $3 \sin \theta + 5 \cos \theta = 5$, then what is the value of $5 \sin \theta - 3 \cos \theta$ equal to?

[CDS-2023 (2)]

(a) 5

(b) -3

(c) -2

(d) 0

234. What is $3(\sin x - \cos)^4 + 6(\sin x + \cos)^2 + 4(\sin x)^6 + 4(\cos x)^6$ equal to?

[CDS-2023 (2)]

(a) 9

(b) 11

(c) 13

(d) 15

235. What is the value of $\sin \theta + \cos \theta$, if θ satisfies the equation $\cot^2 \theta - (\sqrt{3}+1) \cot \theta + \sqrt{3} = 0$; $0 < \theta < \pi/4$?

[CDS-2023 (2)]

(a) $\sqrt{2}$

(b) 2

(c) $\frac{\sqrt{3}+1}{2}$

(d) $\frac{\sqrt{3}-1}{2}$

236. Which one of the following is a value of θ , if θ satisfies the equation $\tan 2\theta \tan 4\theta - 1 = 0$; $0 < \theta < \pi/2$?

[CDS-2023 (2)]

(a) $\pi/12$

(b) $\pi/15$

(c) $\pi/6$

(d) $\pi/5$

237. If $\tan x = \frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta}$, $\frac{\pi}{2} < \theta < \frac{3\pi}{2}$, then what is $\sqrt{2} \sin x$ equal to?

[CDS-2023 (2)]

(a) $\sin \theta + \cos \theta$

(b) $\sin \theta - \cos \theta$

(c) $\frac{\sin \theta + \cos \theta}{2}$

(d) $\frac{\sin \theta - \cos \theta}{2}$

238. How many values of θ will satisfy the equation $(\sin^2 \theta - 4 \sin \theta + 3)(4 - \cos^2 \theta + 4 \sin \theta) = 0$, where $0 < \theta < \pi/2$?

[CDS-2023 (2)]

(a) none

(b) only one

(c) only two

(d) only three

239. If $x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$ and $x \sin \theta - y \cos \theta = 0$, for every $\theta \in \left(0, \frac{\pi}{2}\right)$, then what is $x^2 + y^2$ equal to?

[CDS-2023 (2)]

(a) 0

(b) 1

(c) 2

(d) 3

240. For what relation between a and b is the equation $\sin \theta = \frac{a+b}{2\sqrt{ab}}$ possible?

[CDS-2023 (2)]

(a) $a = b$

(b) $a \leq b$

(c) $a \geq b$

(d) $a > b$

241. If $\tan (3A) = \cot (A - 22^\circ)$ where $3A$ is an acute length, then what is the value of A ?

[CDS-2024 (1)]

(a) 25°

(b) 27°

(c) 28°

(d) 30°

242.

If $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = p \sec \theta + q \tan \theta$,

where $0 < \theta < \frac{\pi}{2}$, then what is $p + q$ equal to?

[CDS-2024 (1)]

(a) 0

(b) 1

(c) 2

(d) 4

243. Consider the following statements

1. In a triangle ABC, if

$\sin A + \sin B + \sin C = \frac{3\sqrt{3}}{2}$, then the

triangle can be equilateral

2. In a triangle ABC, if

$\cos A + \cos B + \cos C = \frac{3}{2}$, then the

triangle can be equilateral.

Which of the statements given below is/are correct?

[CDS-2024 (1)]

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

244. If $3 \sin \theta + 5 \cos \theta = 5$, then what is the value of $5 \sin \theta - 3 \cos \theta$?

[CDS-2024 (1)]

(a) -3

(b) -2

(c) 5

(d) 8

245. What is the minimum value of $\frac{\sin^2 A + 5 \sin A + 1}{\sin A}$ for $0 < A < \frac{\pi}{2}$?

[CDS-2024 (1)]

(a) 3

(b) 5

(c) 7

(d) 9

246. What is the difference between the greatest value and the least value of $\cos^2 \theta + 3 \sin^2 \theta + 2$?

[CDS-2024 (1)]

- (a) 4
(b) 3
(c) 2
(d) 1

247. If $t = \cos 79^\circ$, then what is $\operatorname{cosec} 79^\circ (1 - \cos 79^\circ)$ equal to?

[CDS-2024 (1)]

- (a) $\sqrt{\frac{1+t}{1-t}}$
(b) $\frac{t}{\sqrt{1-t^2}}$
(c) $\sqrt{\frac{1-t^2}{t}}$
(d) $\sqrt{\frac{1-t}{1+t}}$

ANSWER KEY

1.	b	2.	c	3.	a	4.	b	5.	d	6.	c	7.	c	8.	b	9.	c	10.	b
11.	d	12.	c	13.	d	14.	c	15.	a	16.	a	17.	c	18.	b	19.	a	20.	a
21.	b	22.	a	23.	c	24.	c	25.	a	26.	b	27.	c	28.	c	29.	d	30.	d
31.	b	32.	c	33.	a	34.	d	35.	b	36.	a	37.	d	38.	a	39.	c	40.	a
41.	b	42.	d	43.	d	44.	b	45.	d	46.	d	47.	a	48.	b	49.	d	50.	c
51.	d	52.	c	53.	c	54.	c	55.	c	56.	b	57.	b	58.	b	59.	b	60.	c
61.	a	62.	b	63.	d	64.	a	65.	b	66.	b	67.	c	68.	a	69.	b	70.	a
71.	a	72.	b	73.	a	74.	d	75.	c	76.	d	77.	c	78.	b	79.	b	80.	d
81.	c	82.	a	83.	b	84.	c	85.	d	86.	c	87.	d	88.	b	89.	a	90.	d
91.	c	92.	d	93.	b	94.	c	95.	b	96.	a	97.	a	98.	a	99.	c	100.	d
101.	a	102.	c	103.	d	104.	a	105.	a	106.	a	107.	c	108.	a	109.	b	110.	d
111.	b	112.	b	113.	a	114.	a	115.	c	116.	d	117.	a	118.	b	119.	b	120.	b
121.	c	122.	c	123.	c	124.	c	125.	c	126.	d	127.	b	128.	a	129.	b	130.	c
131.	b	132.	b	133.	c	134.	a	135.	b	136.	a	137.	c	138.	a	139.	d	140.	c
141.	b	142.	c	143.	b	144.	a	145.	a	146.	d	147.	b	148.	b	149.	a	150.	c
151.	c	152.	d	153.	a	154.	b	155.	b	156.	d	157.	d	158.	c	159.	b	160.	b
161.	b	162.	d	163.	b	164.	b	165.	b	166.	b	167.	a	168.	a	169.	a	170.	d
171.	d	172.	a	173.	c	174.	c	175.	d	176.	c	177.	b	178.	d	179.	a	180.	a
181.	b	182.	c	183.	a	184.	c	185.	a	186.	a	187.	c	188.	a	189.	c	190.	a
191.	d	192.	c	193.	a	194.	a	195.	c	196.	a	197.	b	198.	c	199.	a	200.	c
201.	a	202.	c	203.	b	204.	a	205.	a	206.	c	207.	d	208.	c	209.	b	210.	b
211.	a	212.	a	213.	b	214.	a	215.	c	216.	a	217.	a	218.	c	219.	c	220.	a
221.	a	222.	d	223.	b	224.	a	225.	b	226.	d	227.	c	228.	b	229.	d	230.	c
231.	d	232.	c	233.	b	234.	c	235.	c	236.	a	237.	a	238.	c	239.	b	240.	a
241.	c	242.	c	243.	c	244.	a	245.	c	246.	c	247.	d						